

Episteme Benchmark Executive Summary

Precision Performance Analytics | Generated: 2026-03-12 19:14:30

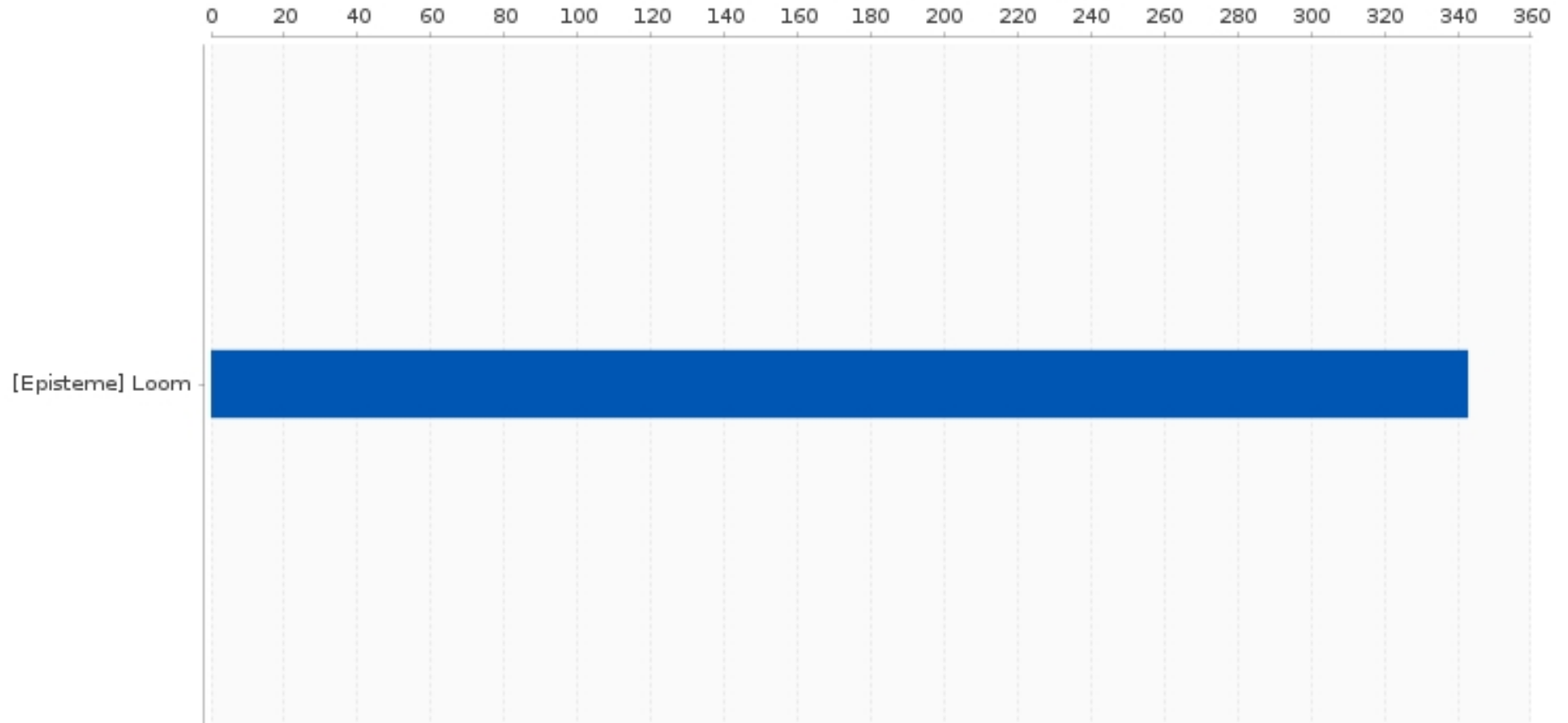
Operating System	Linux (amd64)
Java Runtime	25.0.2 (Oracle Corporation)
Logical Processors	4

AI / Multi-Agent Systems

Audio

AI / Multi-Agent Systems | Scaling Performance

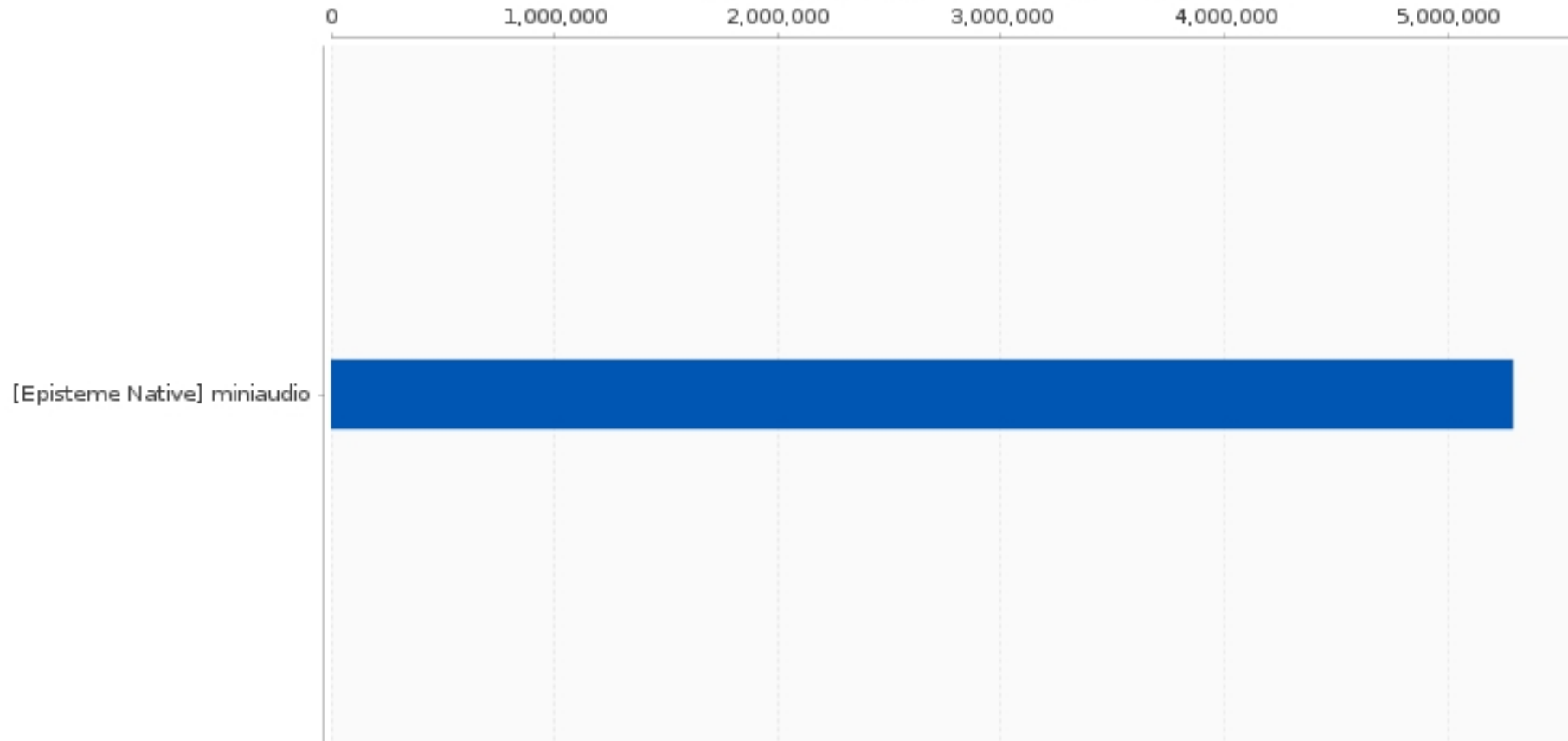
Throughput (Operations per Second)



Bayesian Inference

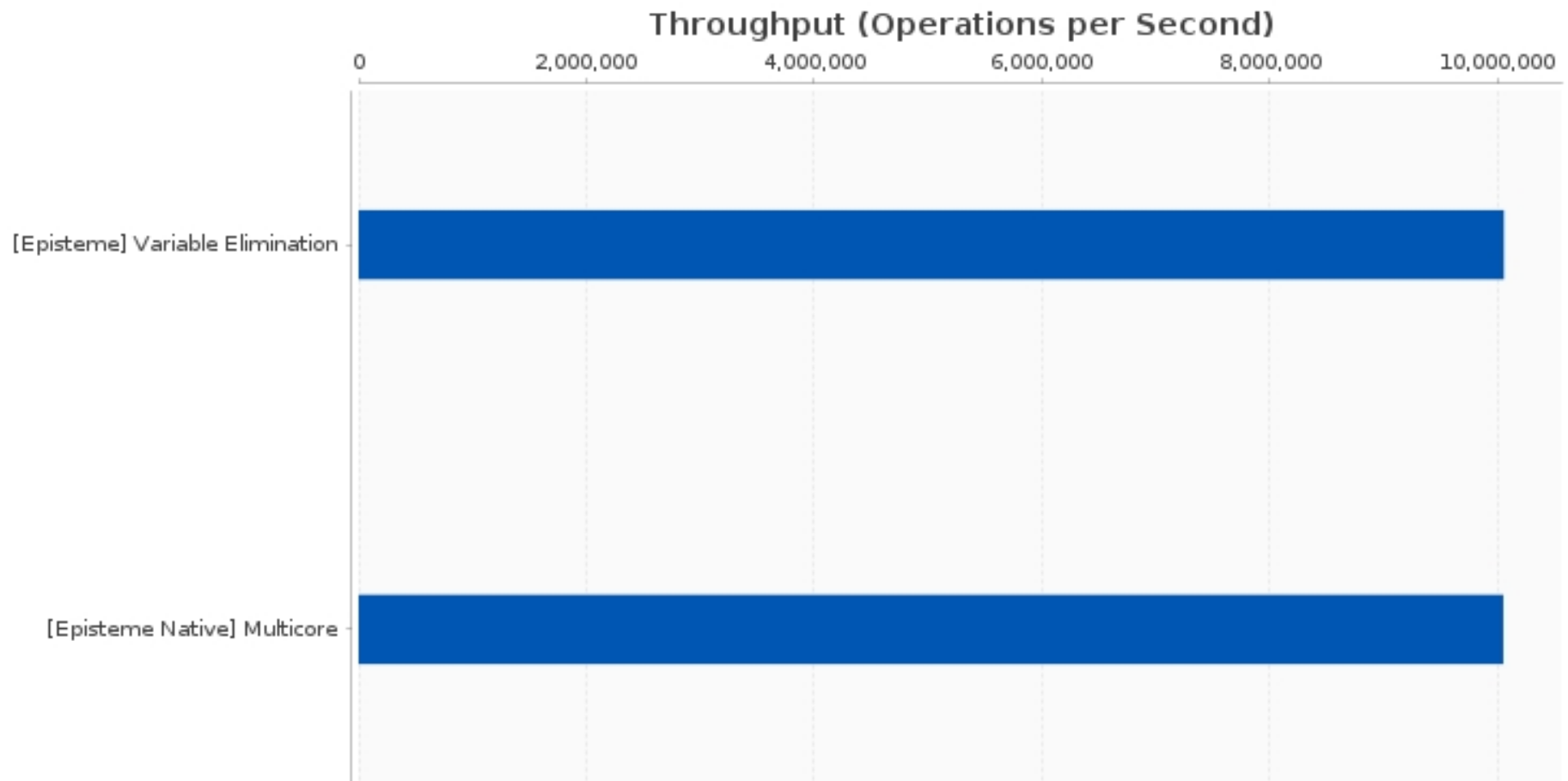
Audio | Scaling Performance

Throughput (Operations per Second)



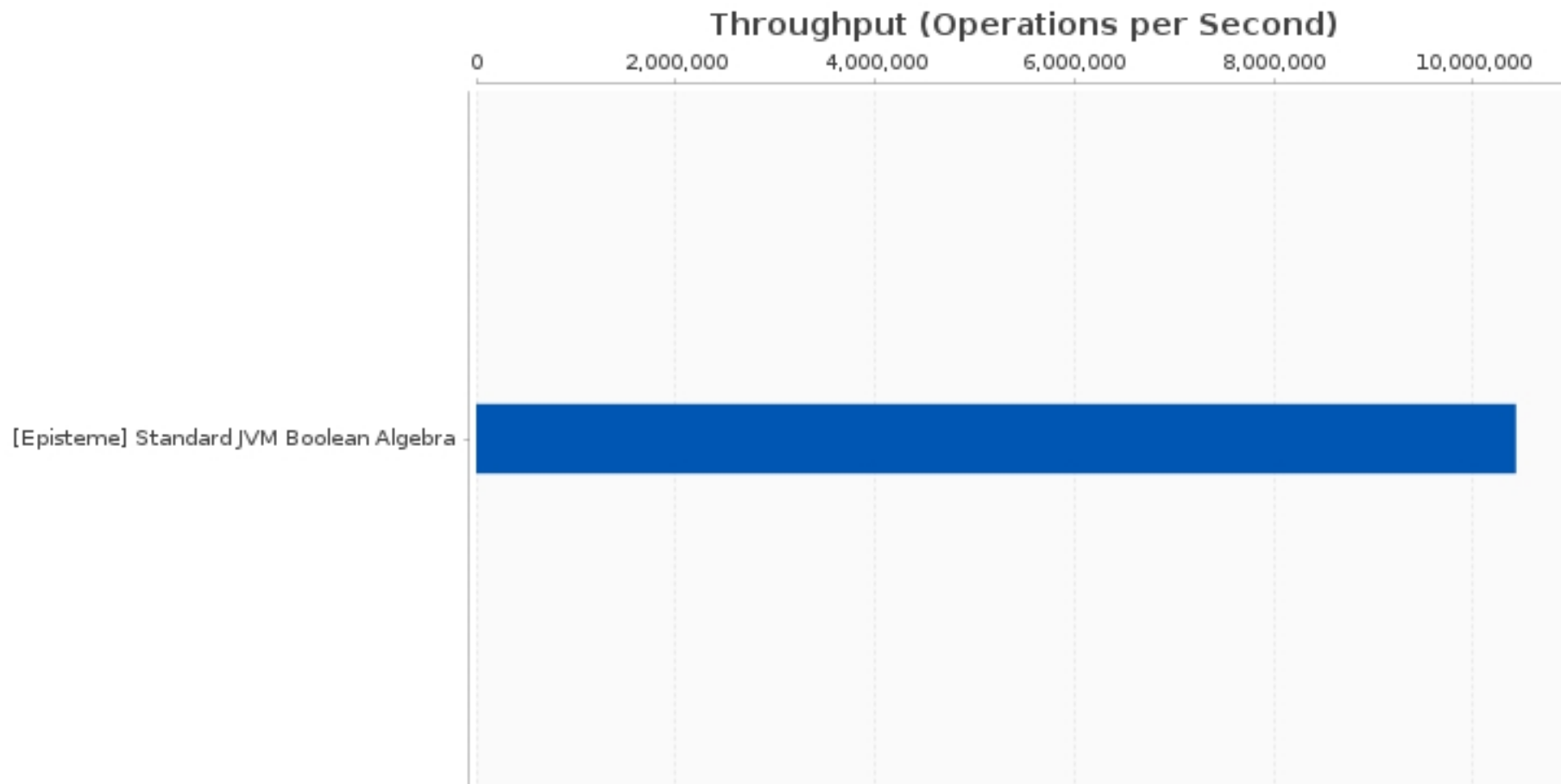
Boolean Algebra

Bayesian Inference | Scaling Performance



Computational Fluid Dynamics

Boolean Algebra | Scaling Performance



Distributed Computing

Computational Fluid Dynamics | Scaling Performance

Throughput (Operations per Second)

0 100 200 300 400 500 600 700 800 900 1,000 1,100 1,200 1,300 1,400 1,500

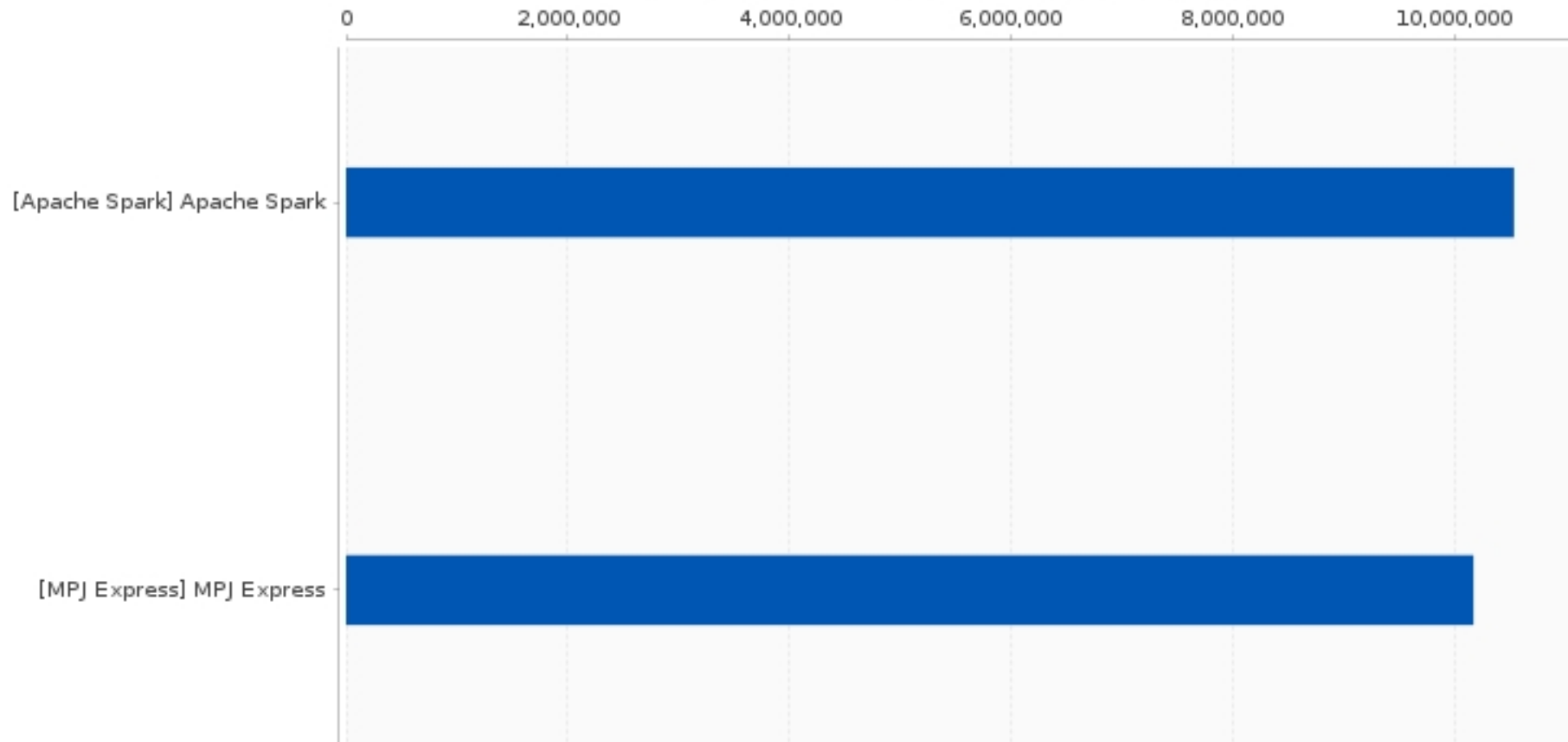
[Episteme Native] LBM GPU



Finite Element Method

Distributed Computing | Scaling Performance

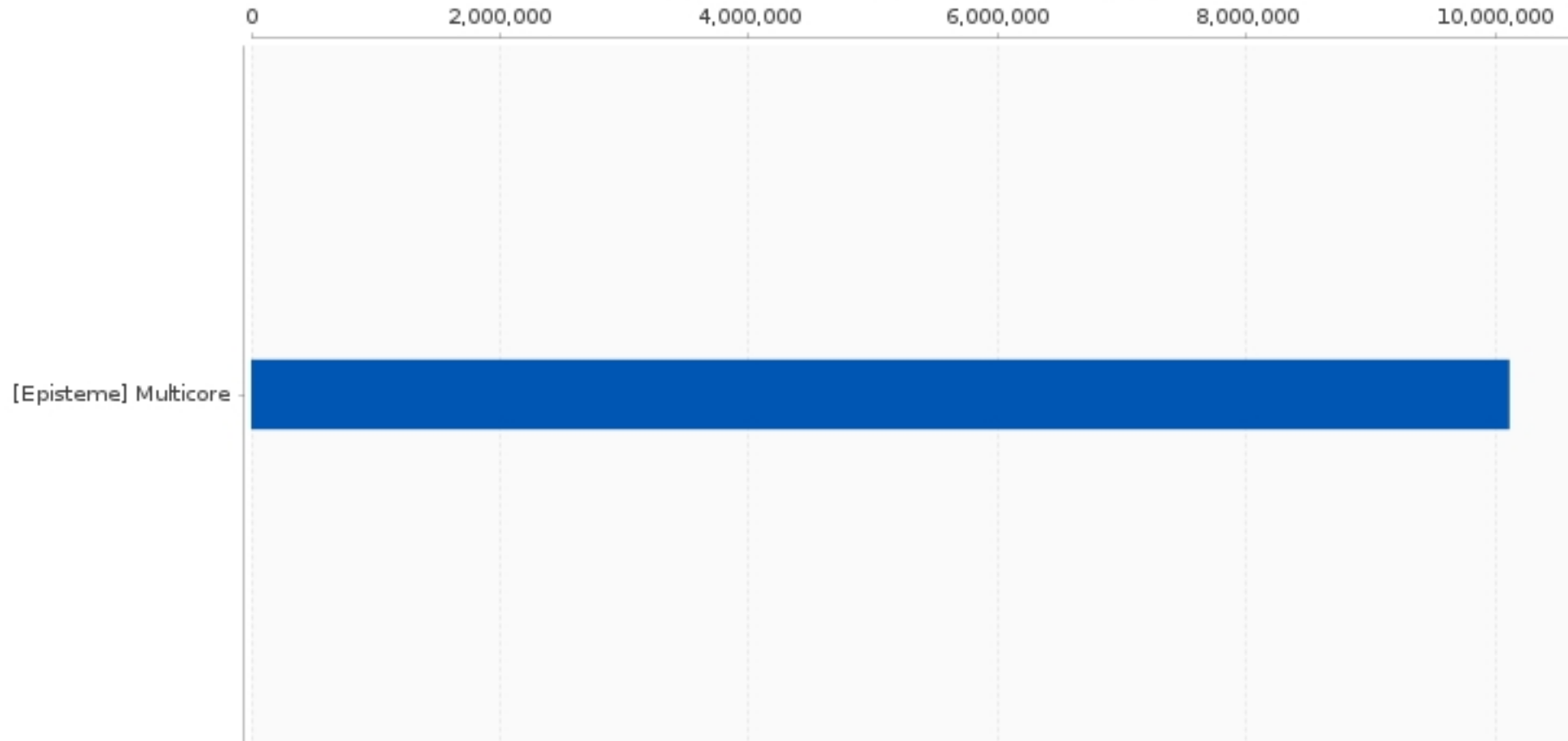
Throughput (Operations per Second)



Fuzzy logic

Finite Element Method | Scaling Performance

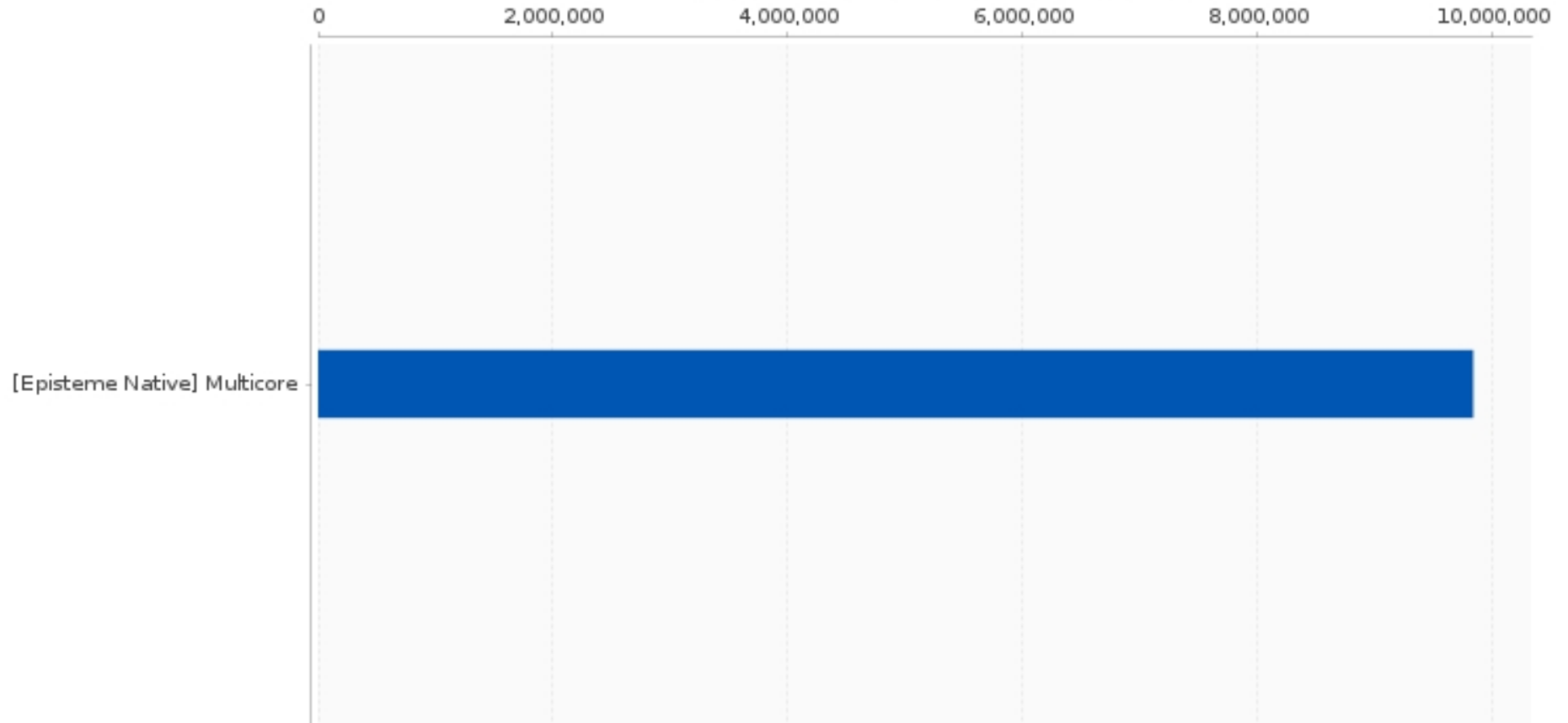
Throughput (Operations per Second)



Genetic Algorithm

Fuzzy logic | Scaling Performance

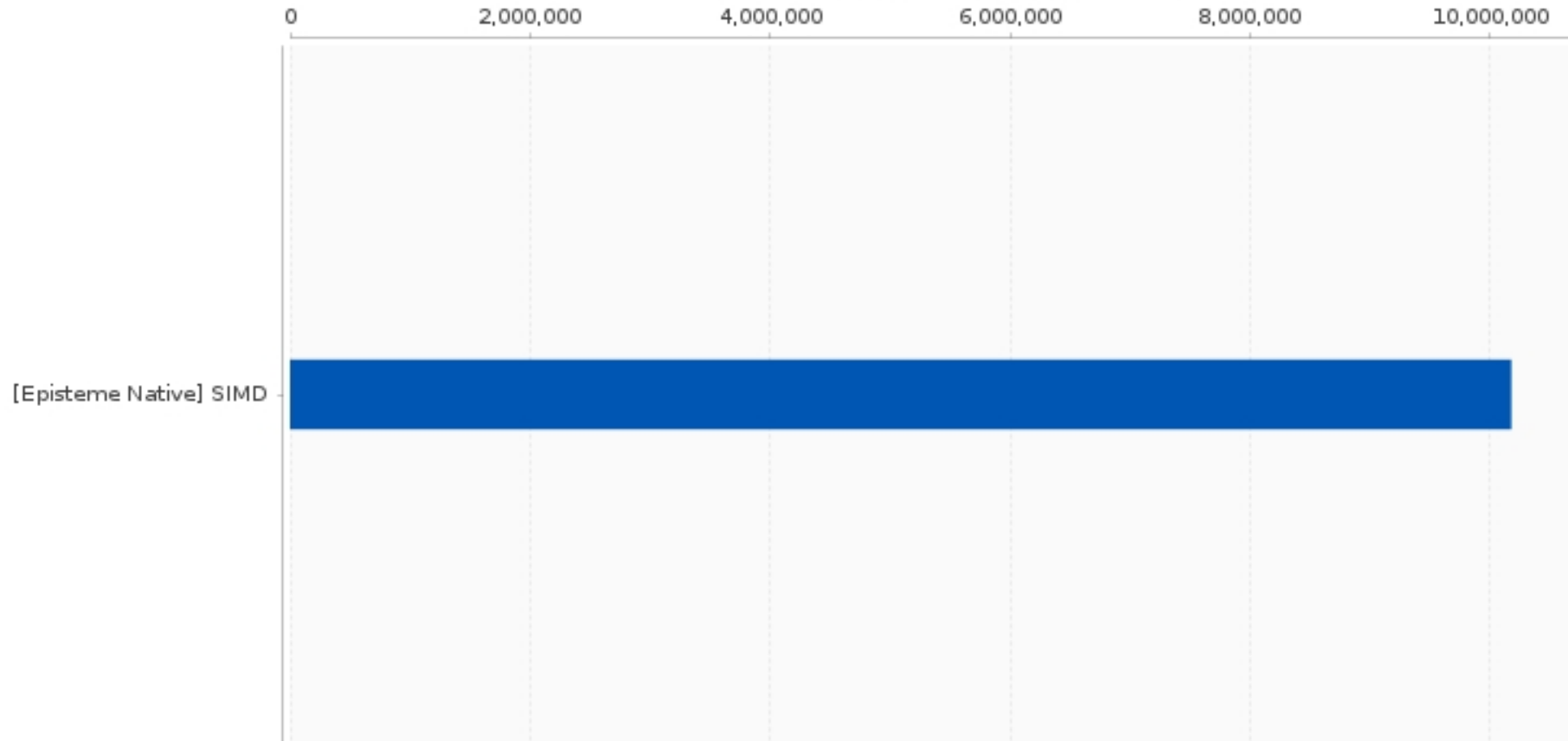
Throughput (Operations per Second)



Graph Algorithm

Genetic Algorithm | Scaling Performance

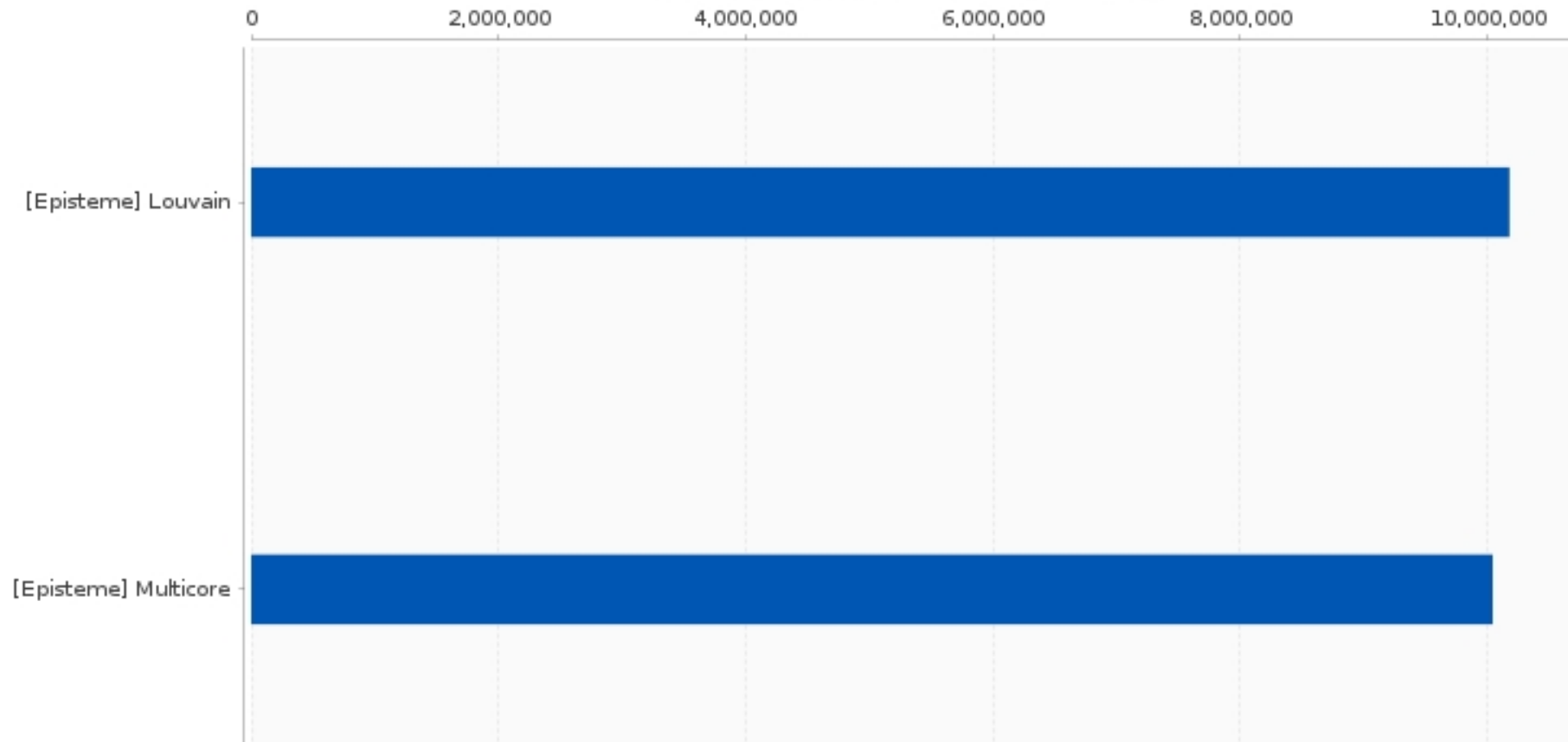
Throughput (Operations per Second)



Input/Output

Graph Algorithm | Scaling Performance

Throughput (Operations per Second)



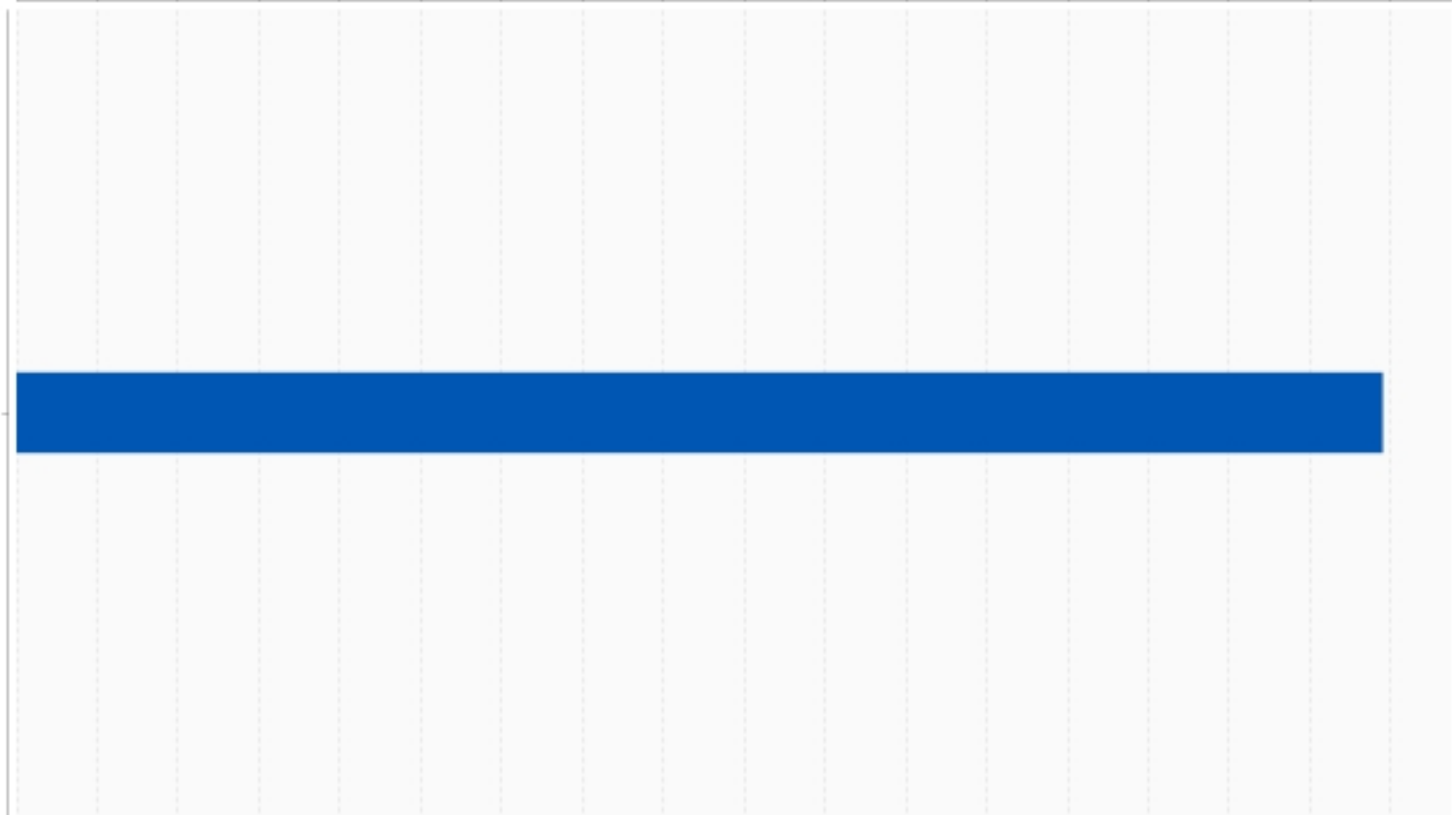
Lattice Boltzmann

Input/Output | Scaling Performance

Throughput (Operations per Second)

0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0 1.1 1.2 1.3 1.4 1.5 1.6 1.7

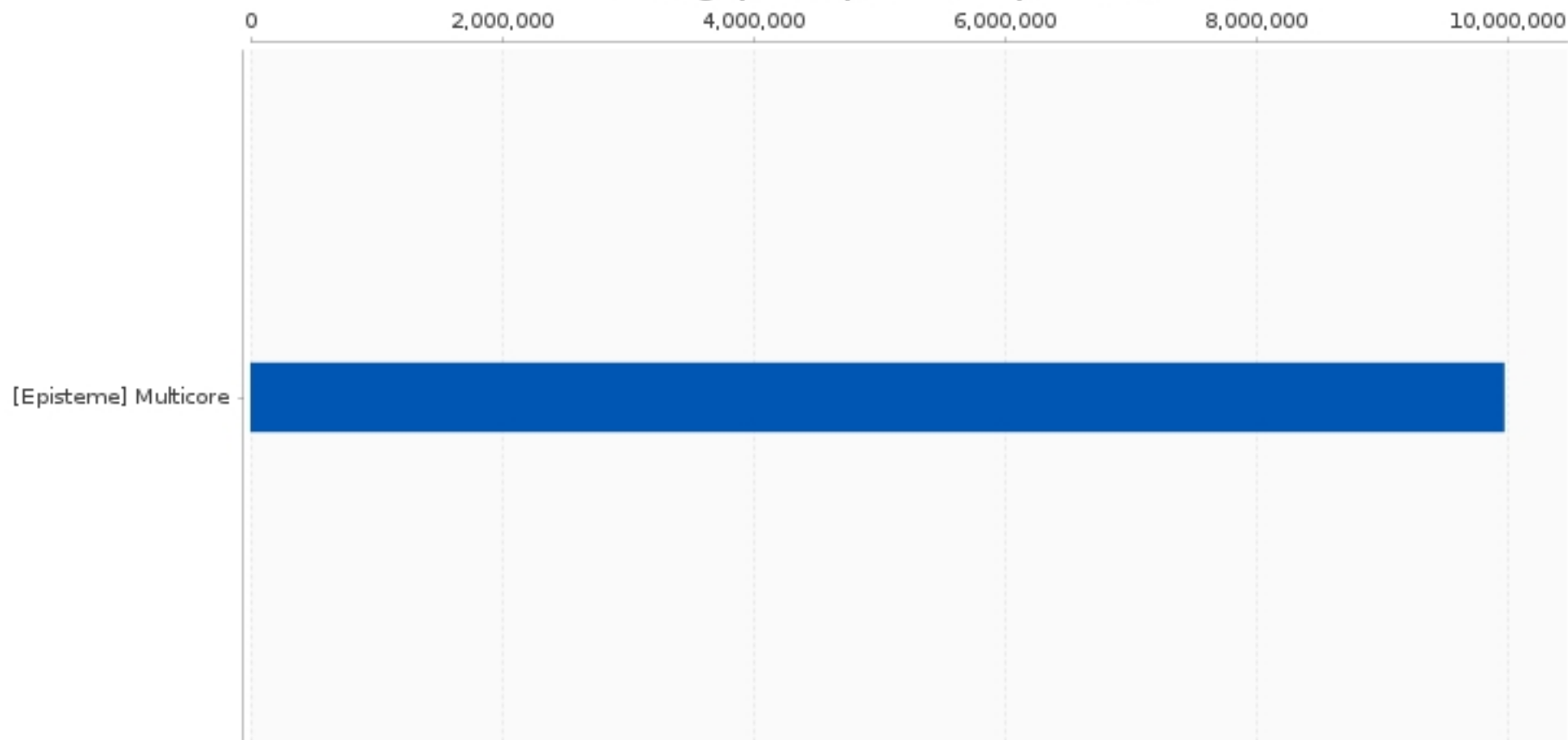
[Episteme Native] Bytedeco



Linear Algebra

Lattice Boltzmann | Scaling Performance

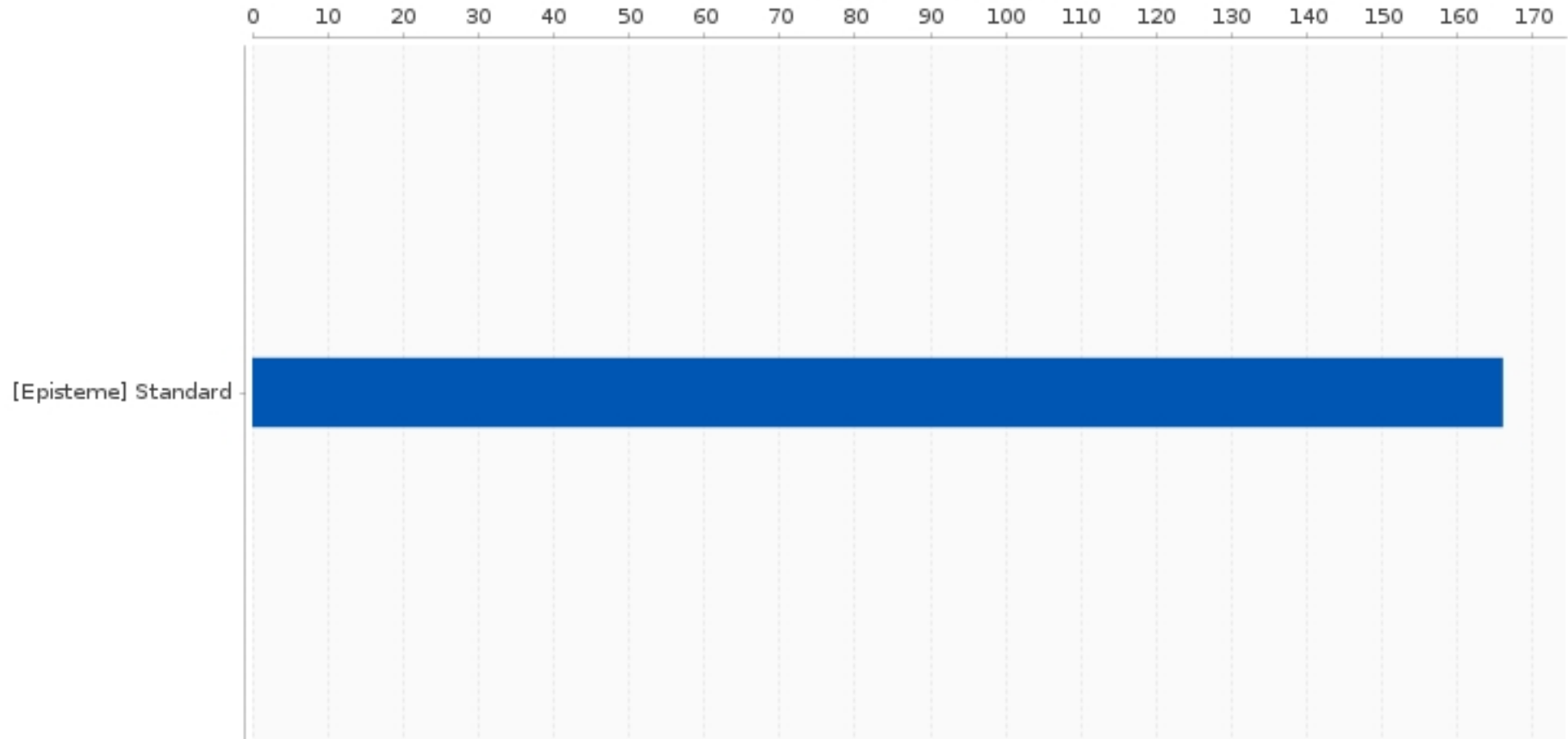
Throughput (Operations per Second)



Linear Algebra (Dense)

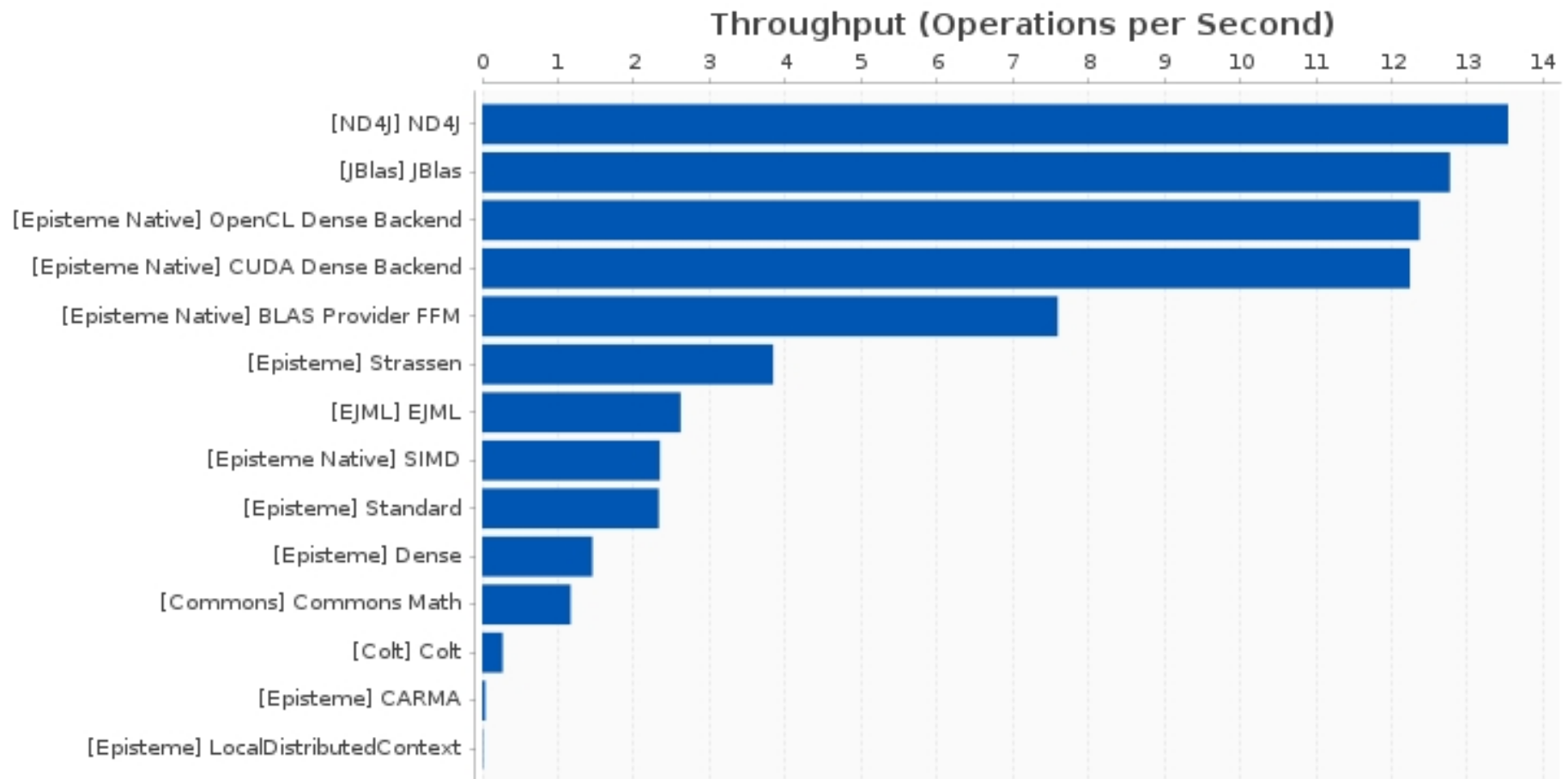
Linear Algebra | Scaling Performance

Throughput (Operations per Second)



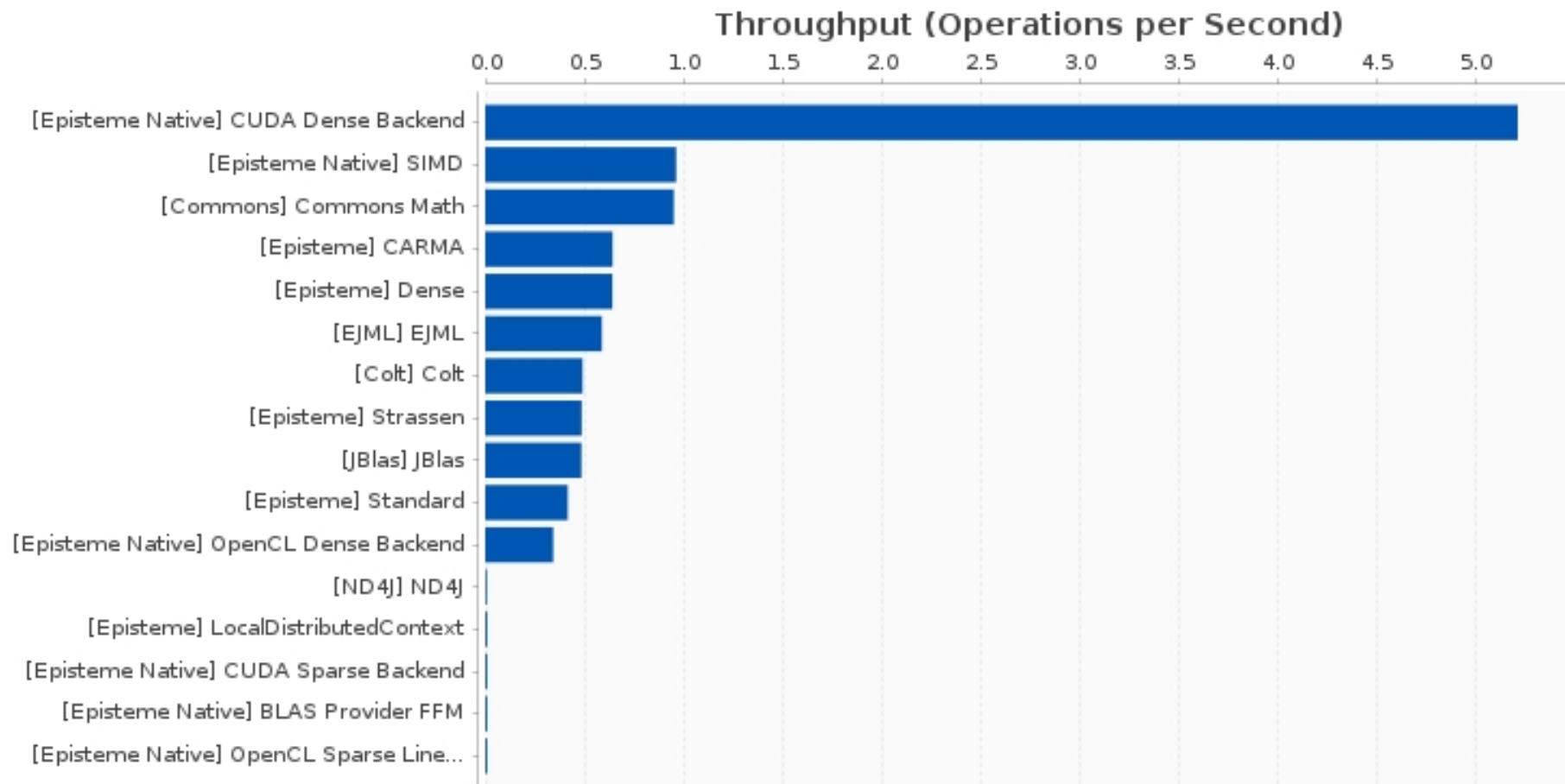
Linear Algebra (Inverse)

Linear Algebra (Dense) | Scaling Performance



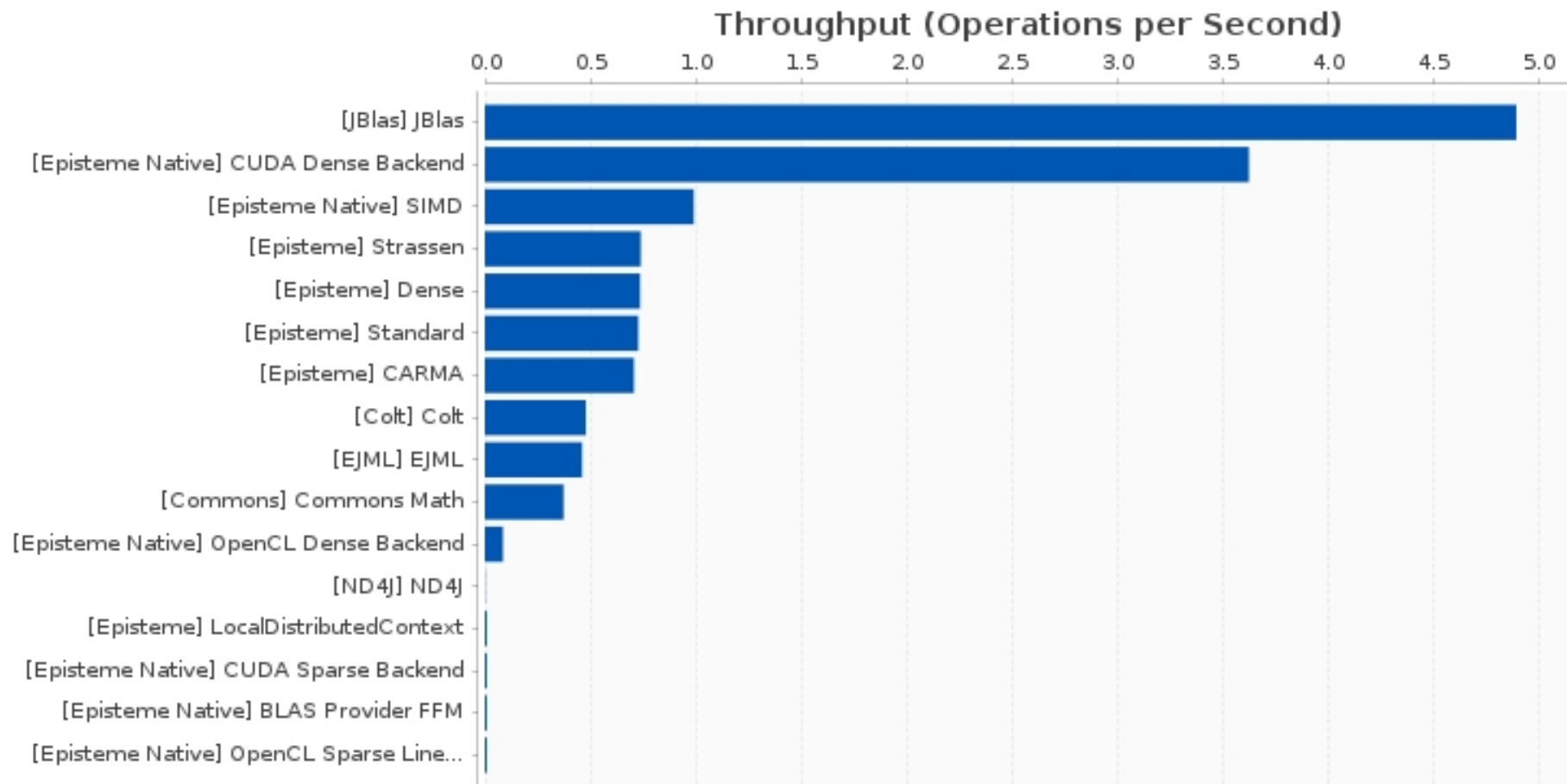
Linear Algebra (Solve)

Linear Algebra (Inverse) | Scaling Performance



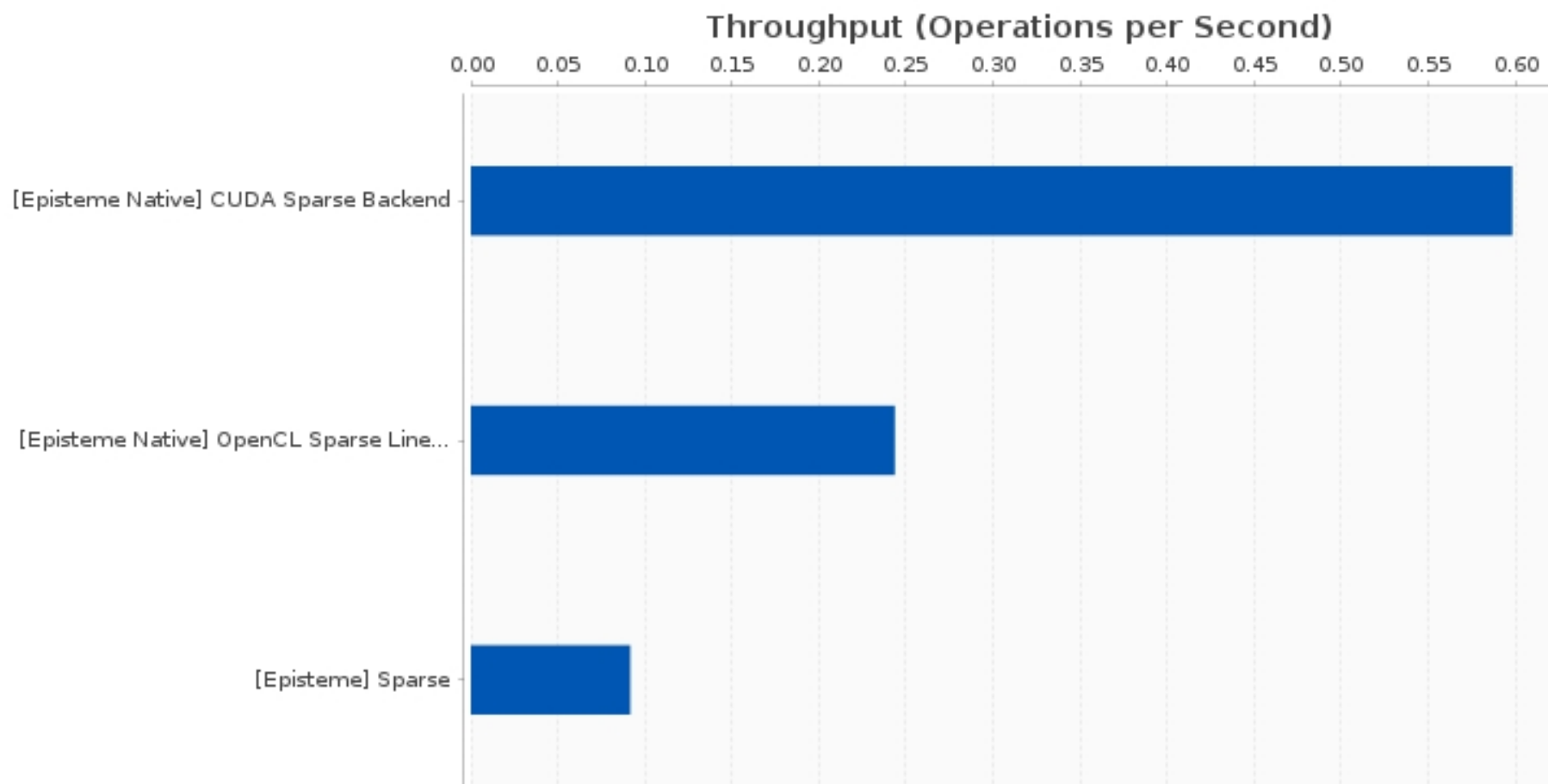
Linear Algebra (Sparse)

Linear Algebra (Solve) | Scaling Performance



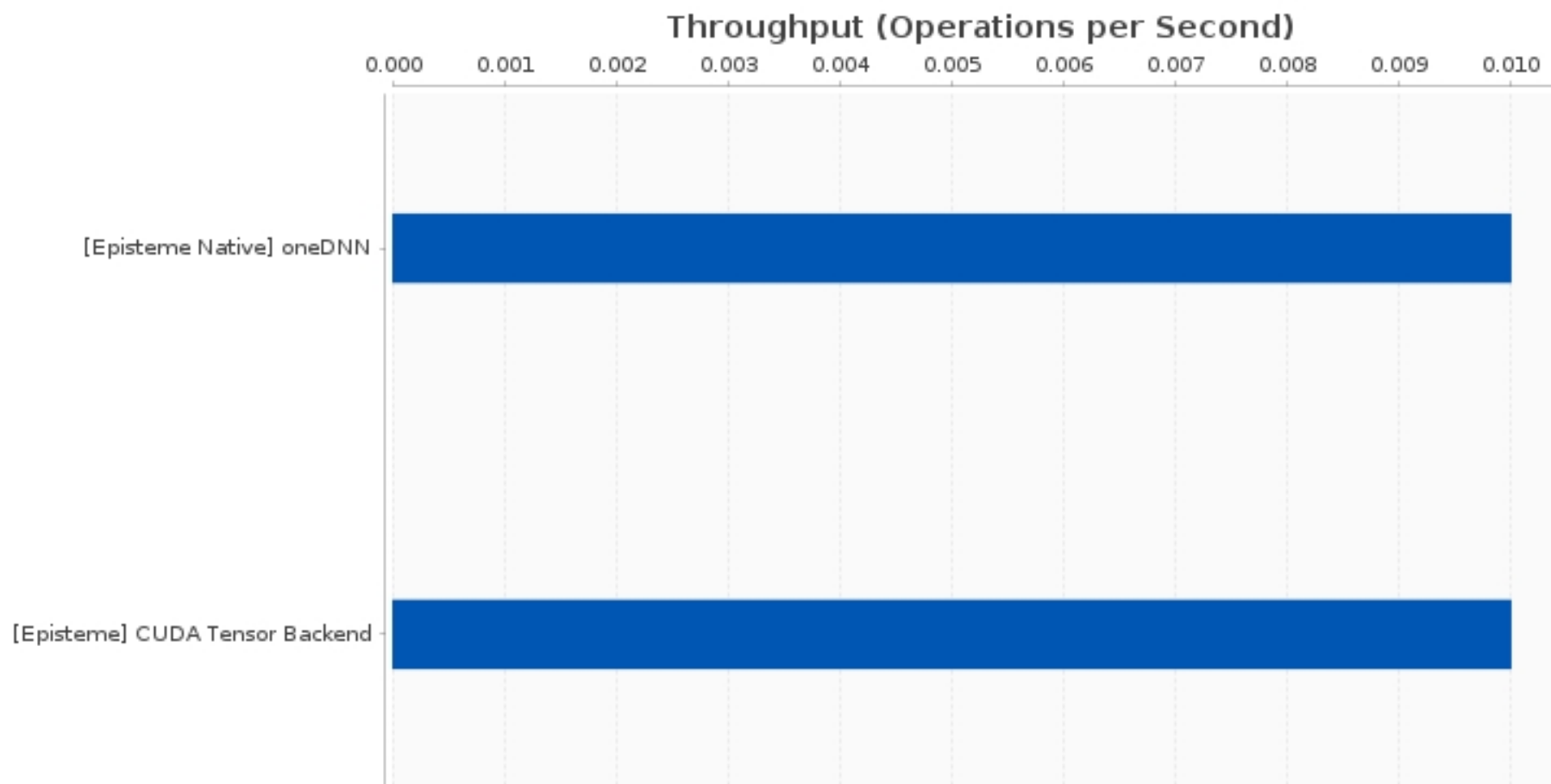
Linear Algebra (Tensor)

Linear Algebra (Sparse) | Scaling Performance



Machine Learning

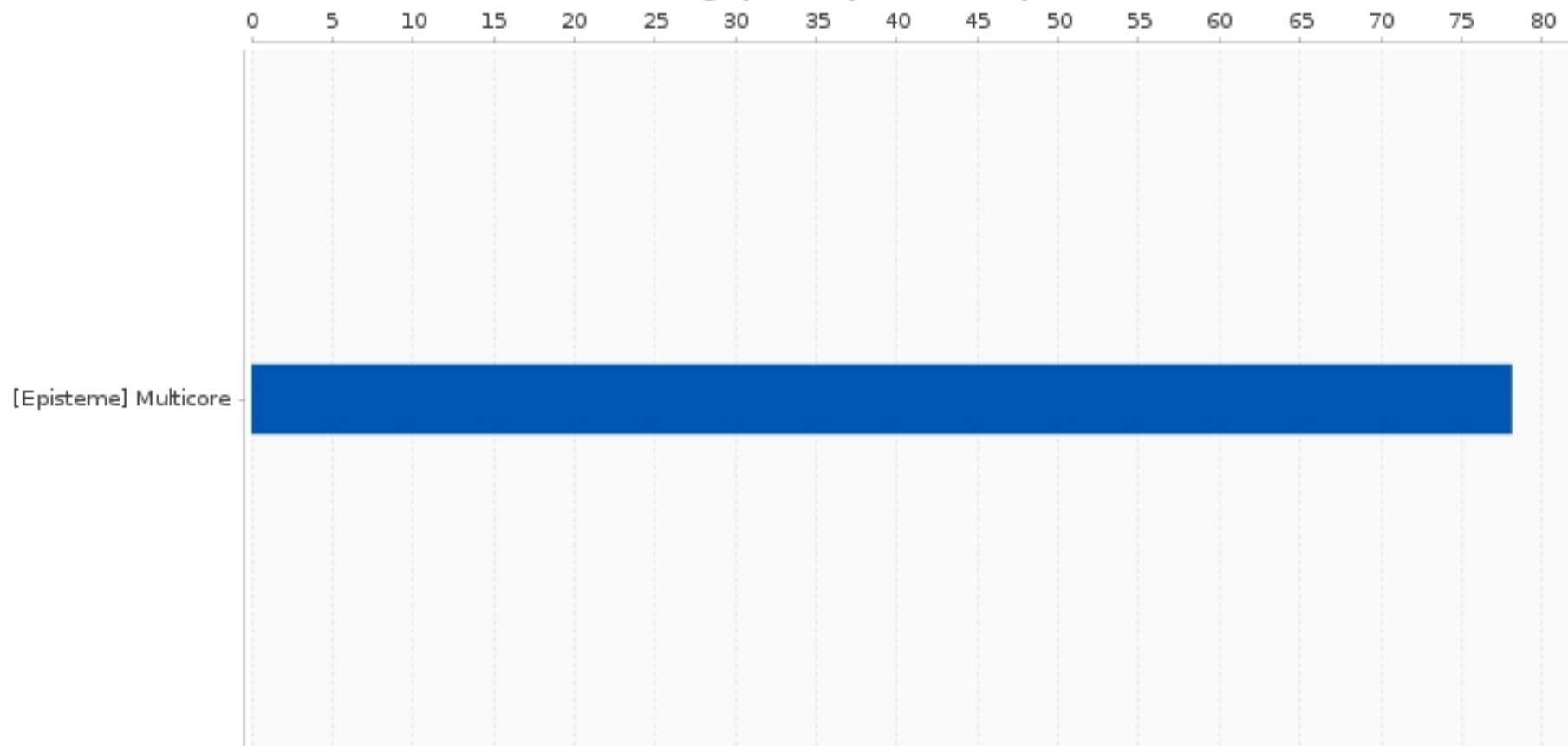
Linear Algebra (Tensor) | Scaling Performance



Machine Learning (ONNX)

Machine Learning | Scaling Performance

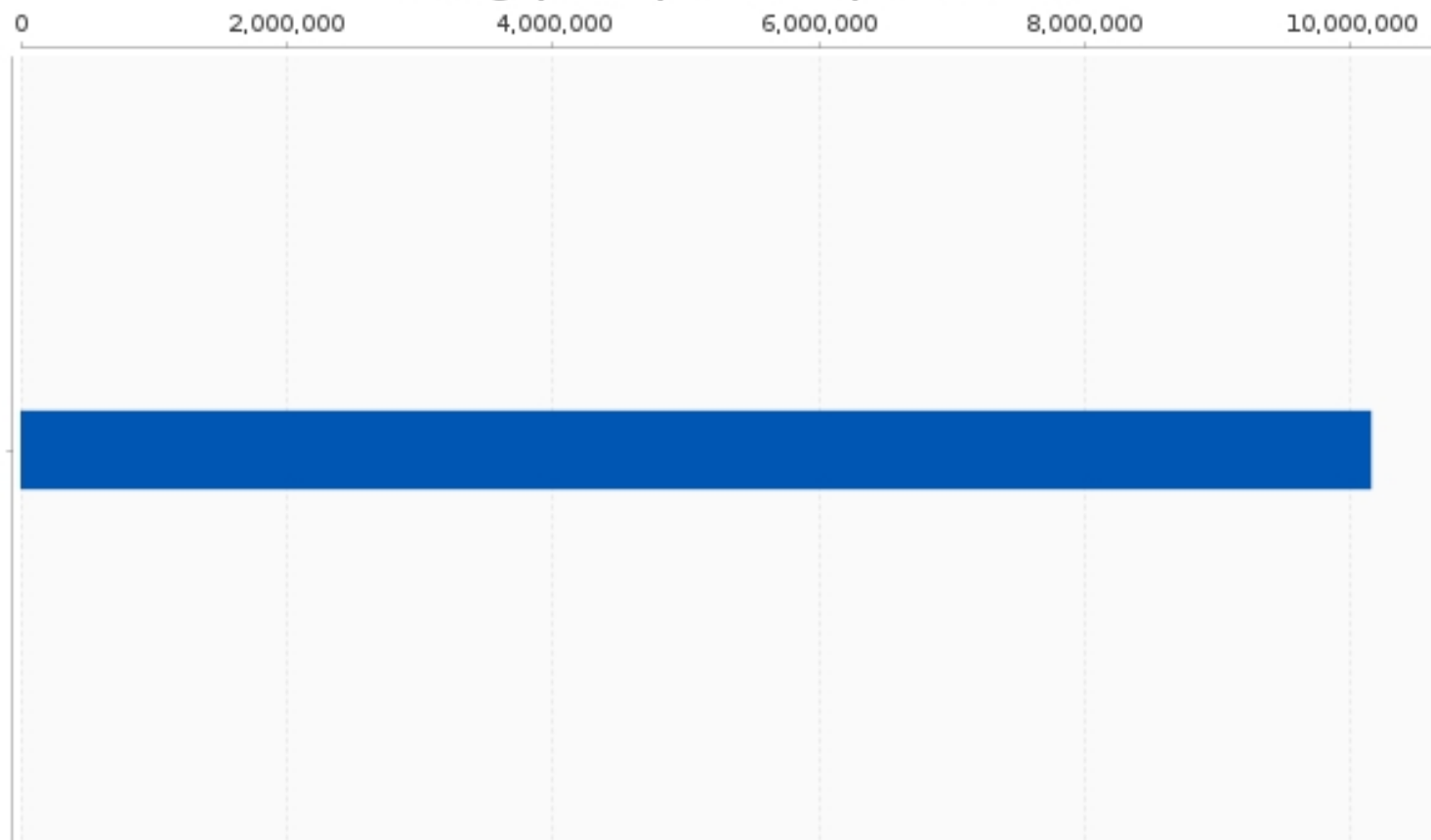
Throughput (Operations per Second)



Maxwell Equations

Machine Learning (ONNX) | Scaling Performance

Throughput (Operations per Second)

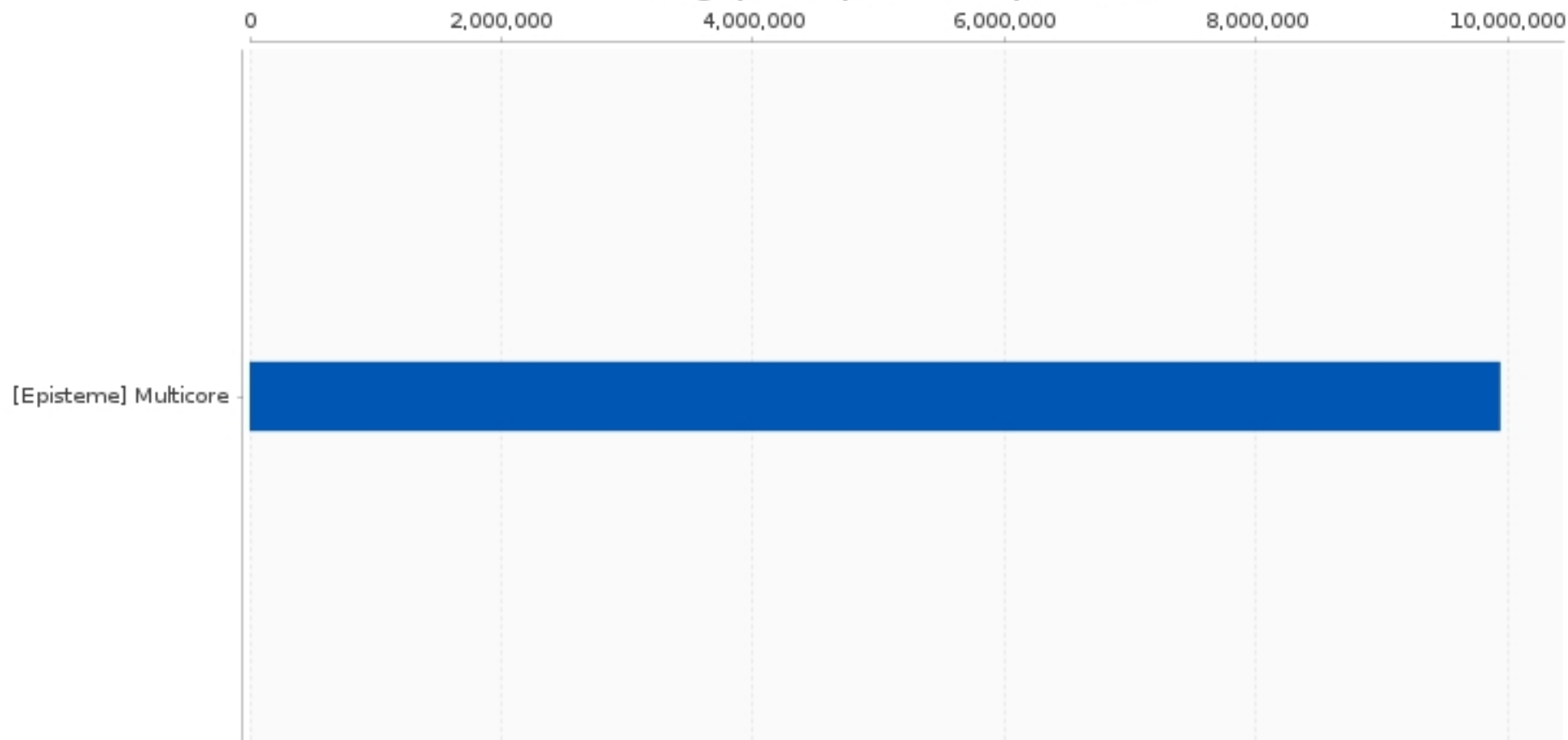


[Episteme] ONNX Runtime

Molecular Dynamics

Maxwell Equations | Scaling Performance

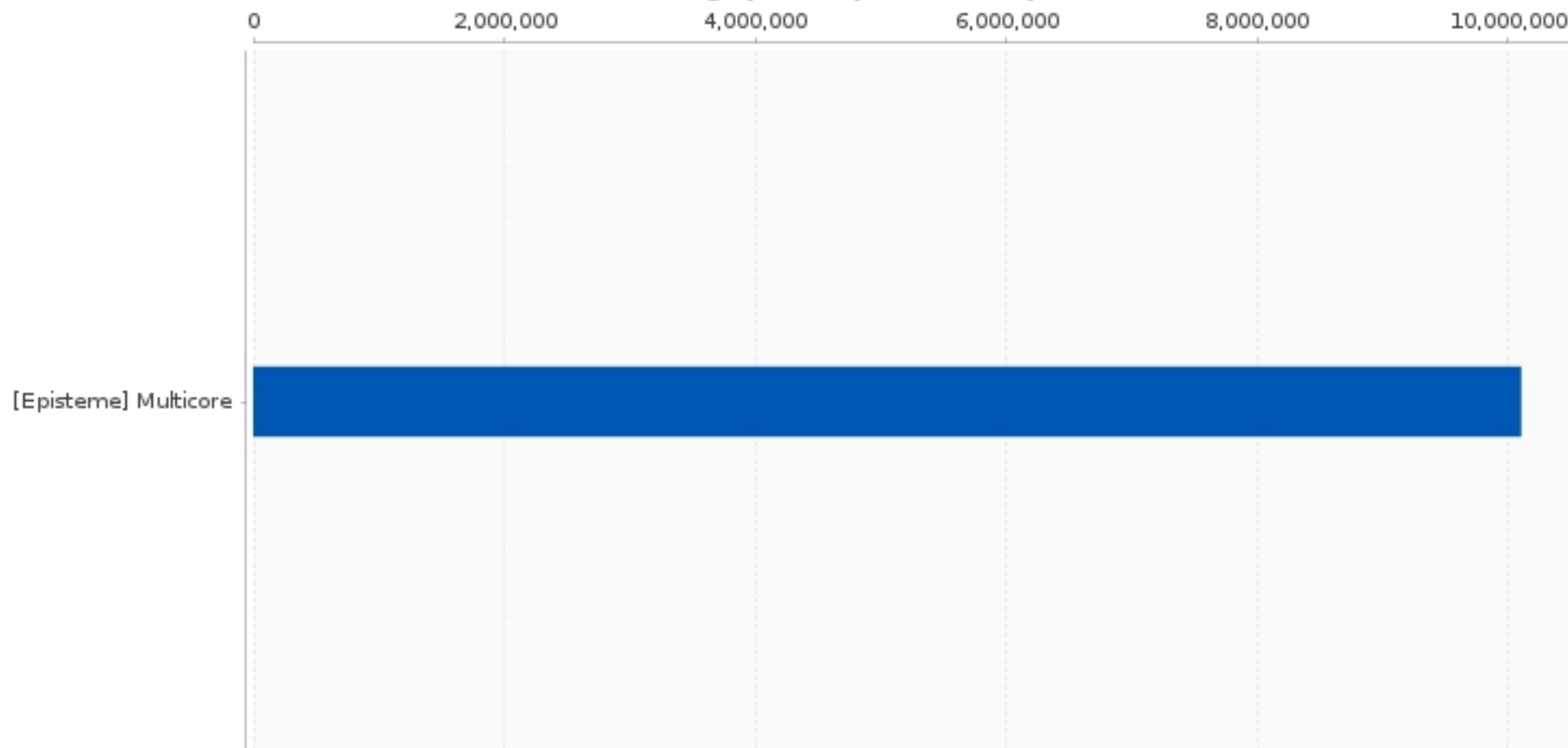
Throughput (Operations per Second)



Monte Carlo Simulation

Molecular Dynamics | Scaling Performance

Throughput (Operations per Second)



N-Body Simulation

Monte Carlo Simulation | Scaling Performance

Throughput (Operations per Second)

0 2,000,000 4,000,000 6,000,000 8,000,000 10,000,000

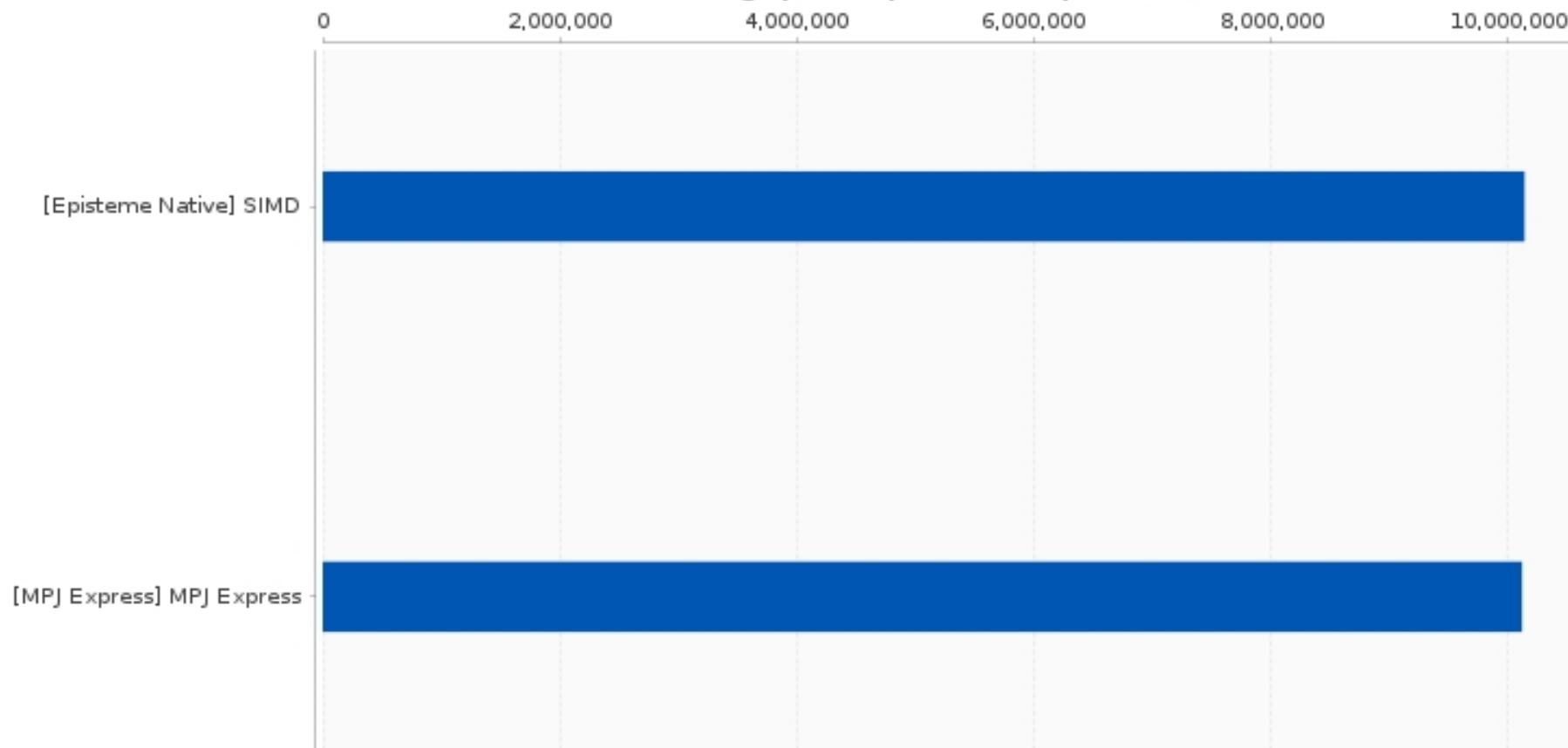
[Episteme Native] Multicore



Navier-Stokes

N-Body Simulation | Scaling Performance

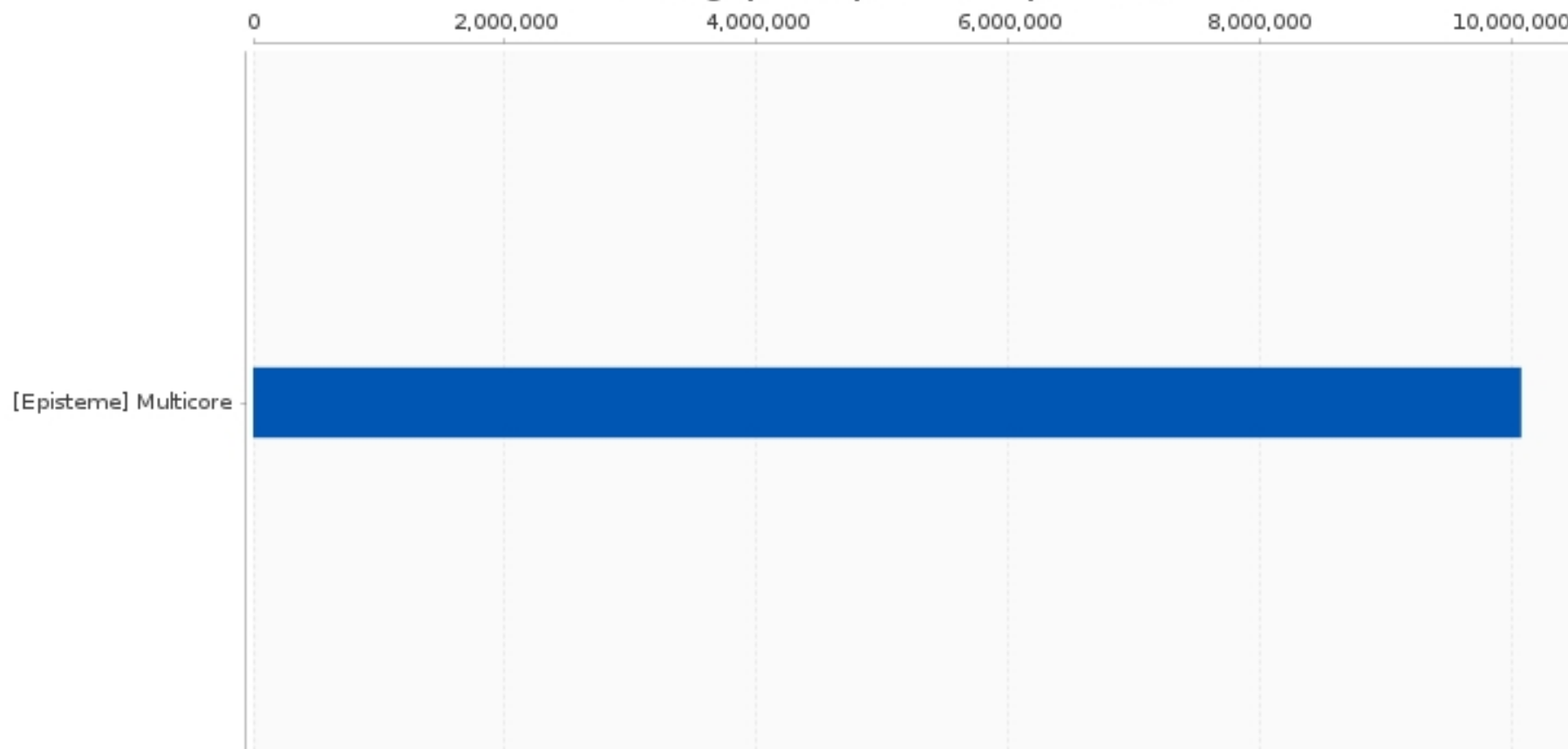
Throughput (Operations per Second)



Neural Inference (ONNX)

Navier-Stokes | Scaling Performance

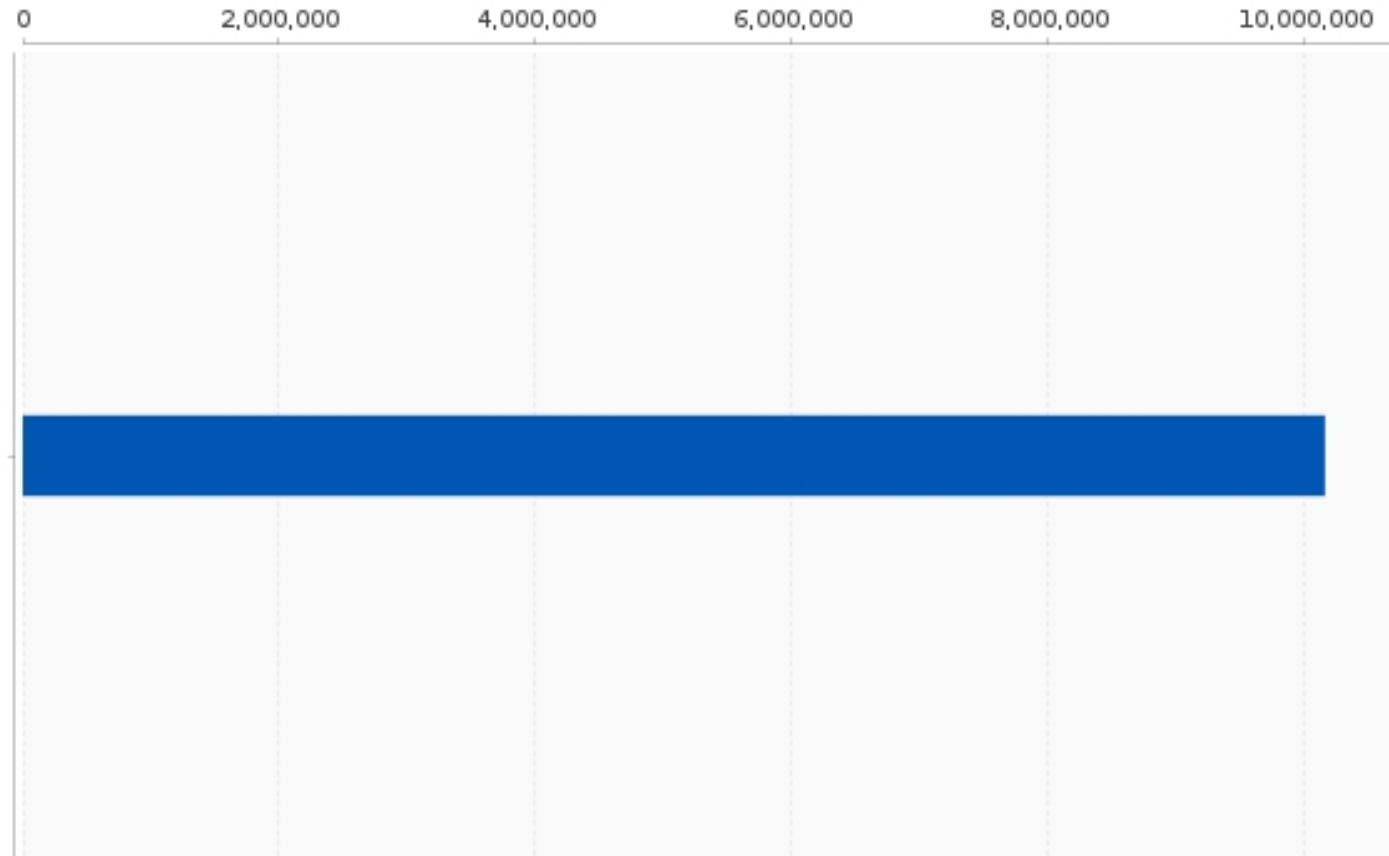
Throughput (Operations per Second)



Neural Operations

Neural Inference (ONNX) | Scaling Performance

Throughput (Operations per Second)

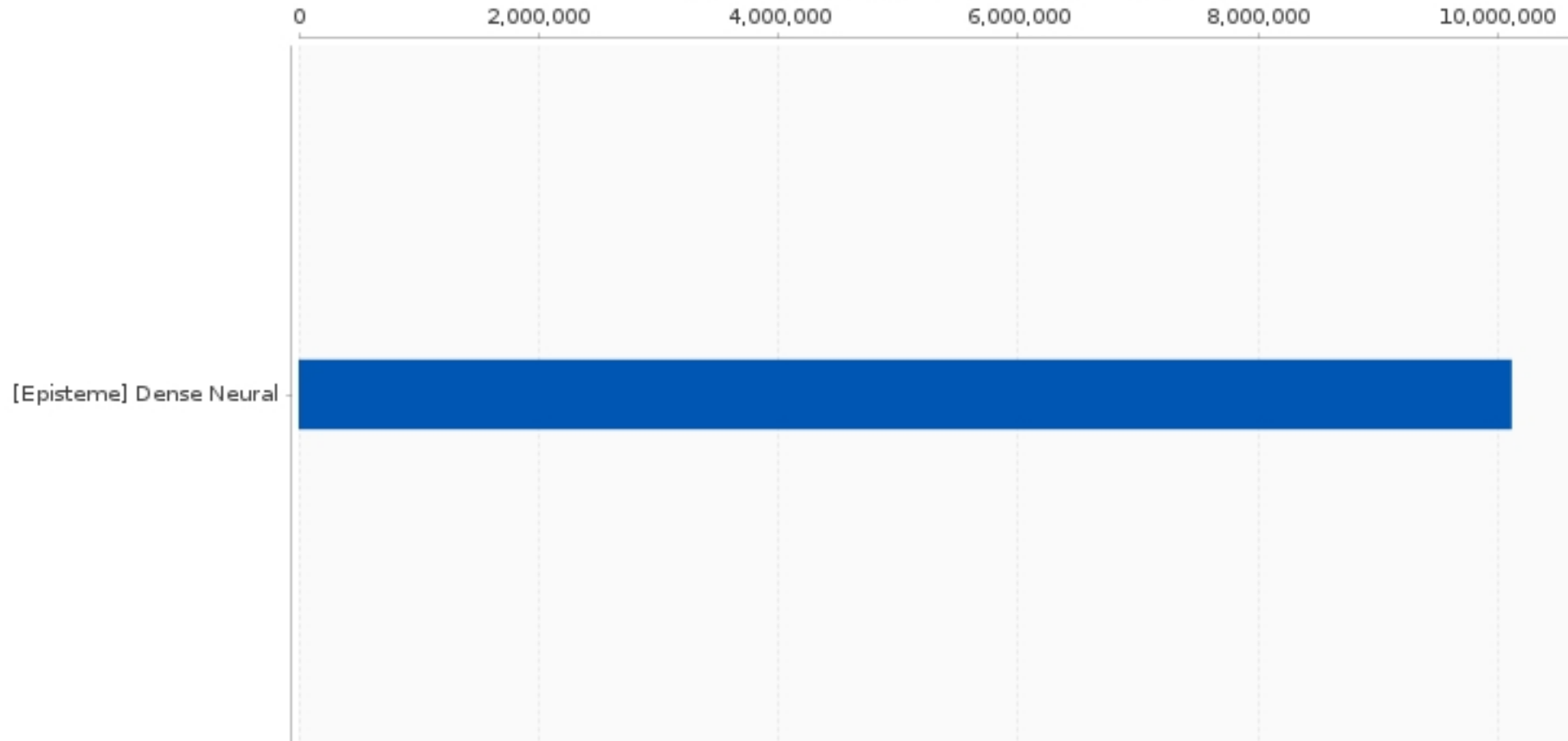


[Episteme] ONNX Runtime Bridge

ODE Solver

Neural Operations | Scaling Performance

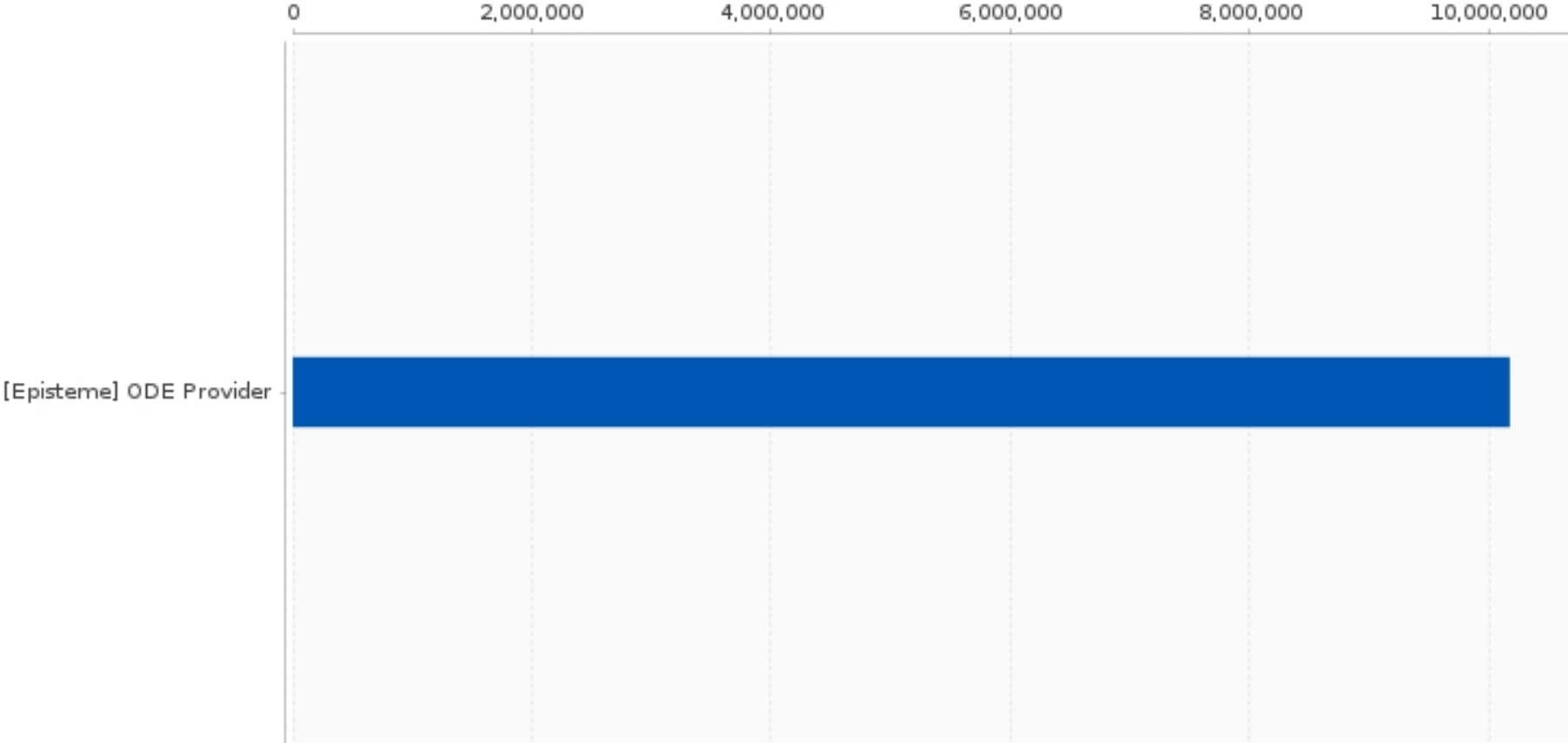
Throughput (Operations per Second)



Parallel Computing

ODE Solver | Scaling Performance

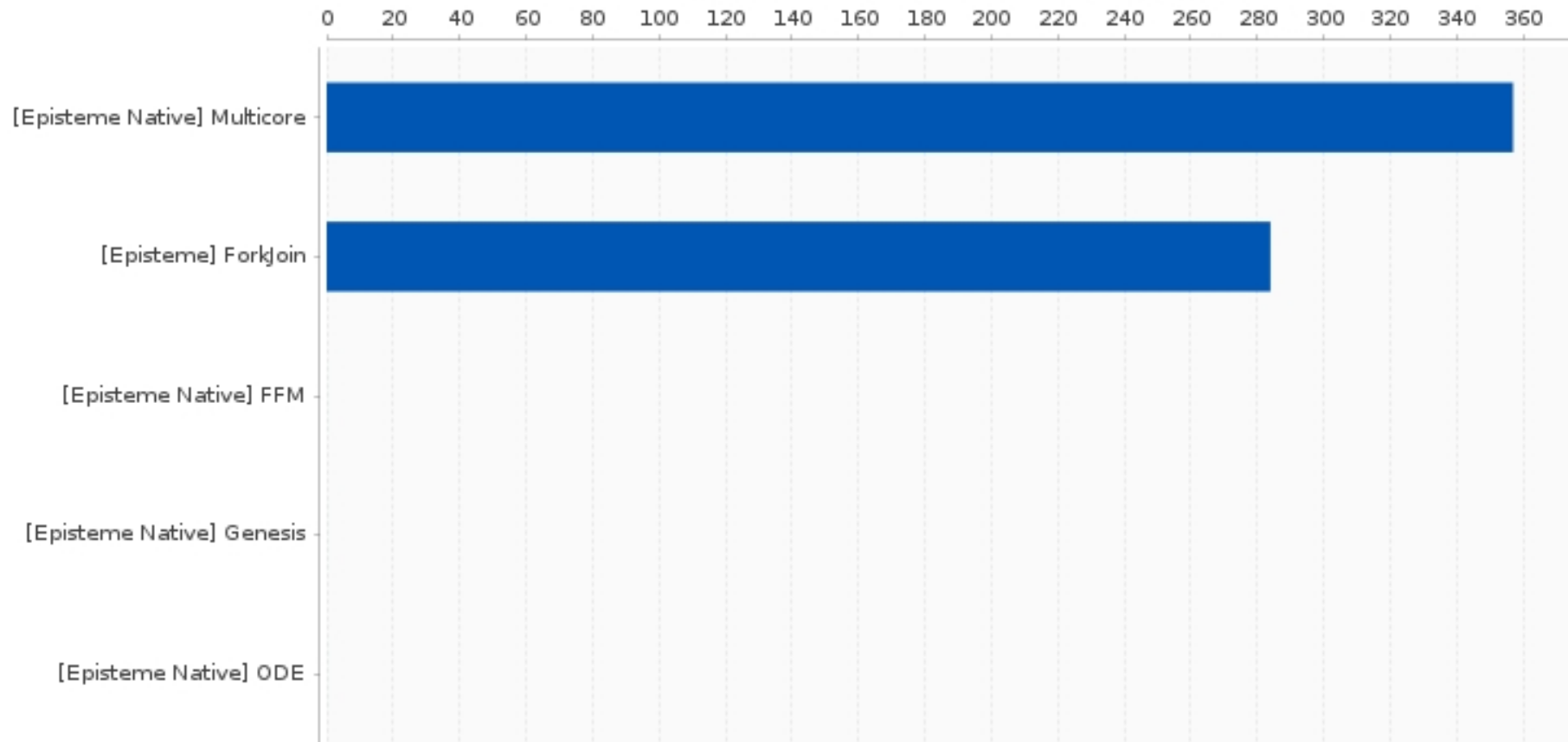
Throughput (Operations per Second)



Physics

Parallel Computing | Scaling Performance

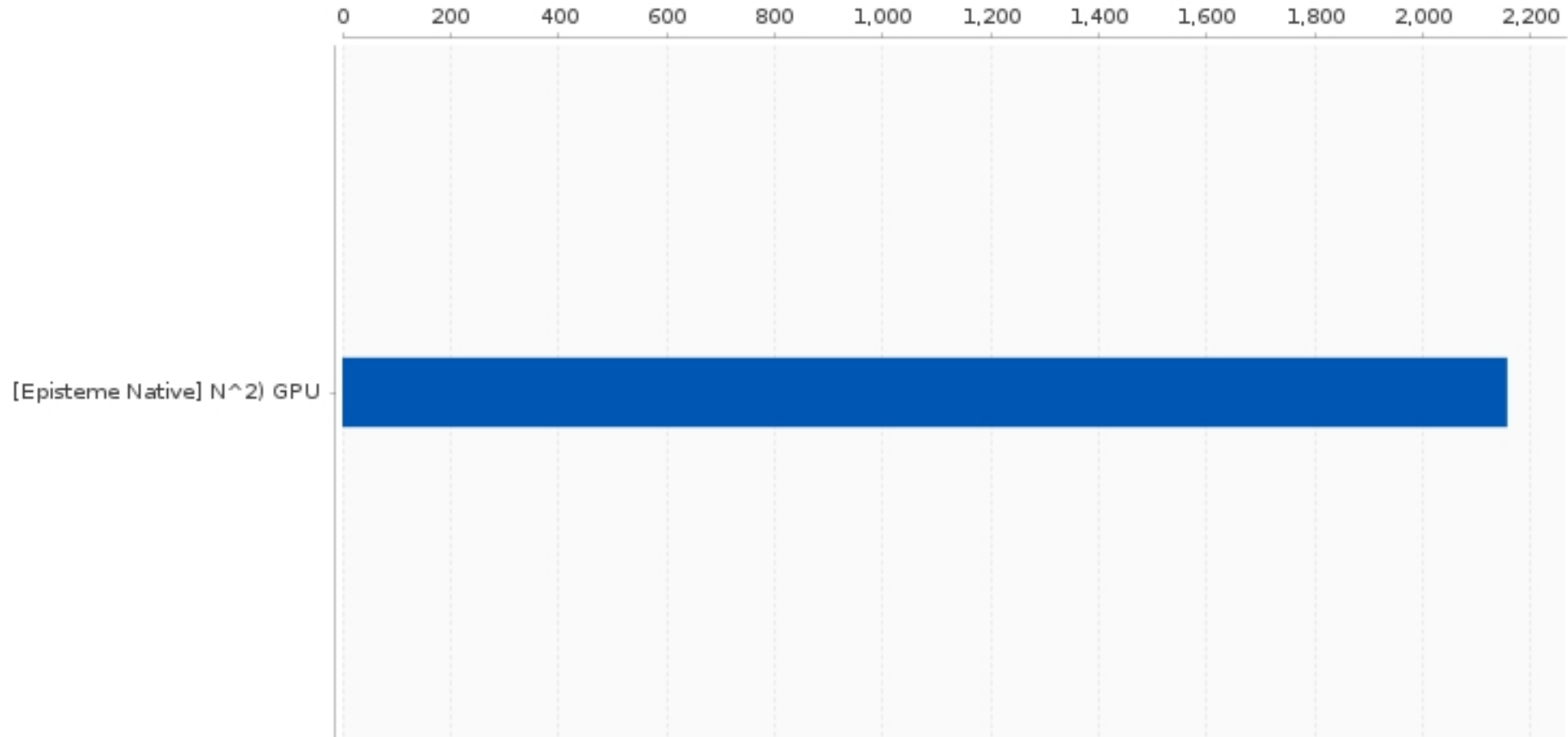
Throughput (Operations per Second)



PhysicsEngine

Physics | Scaling Performance

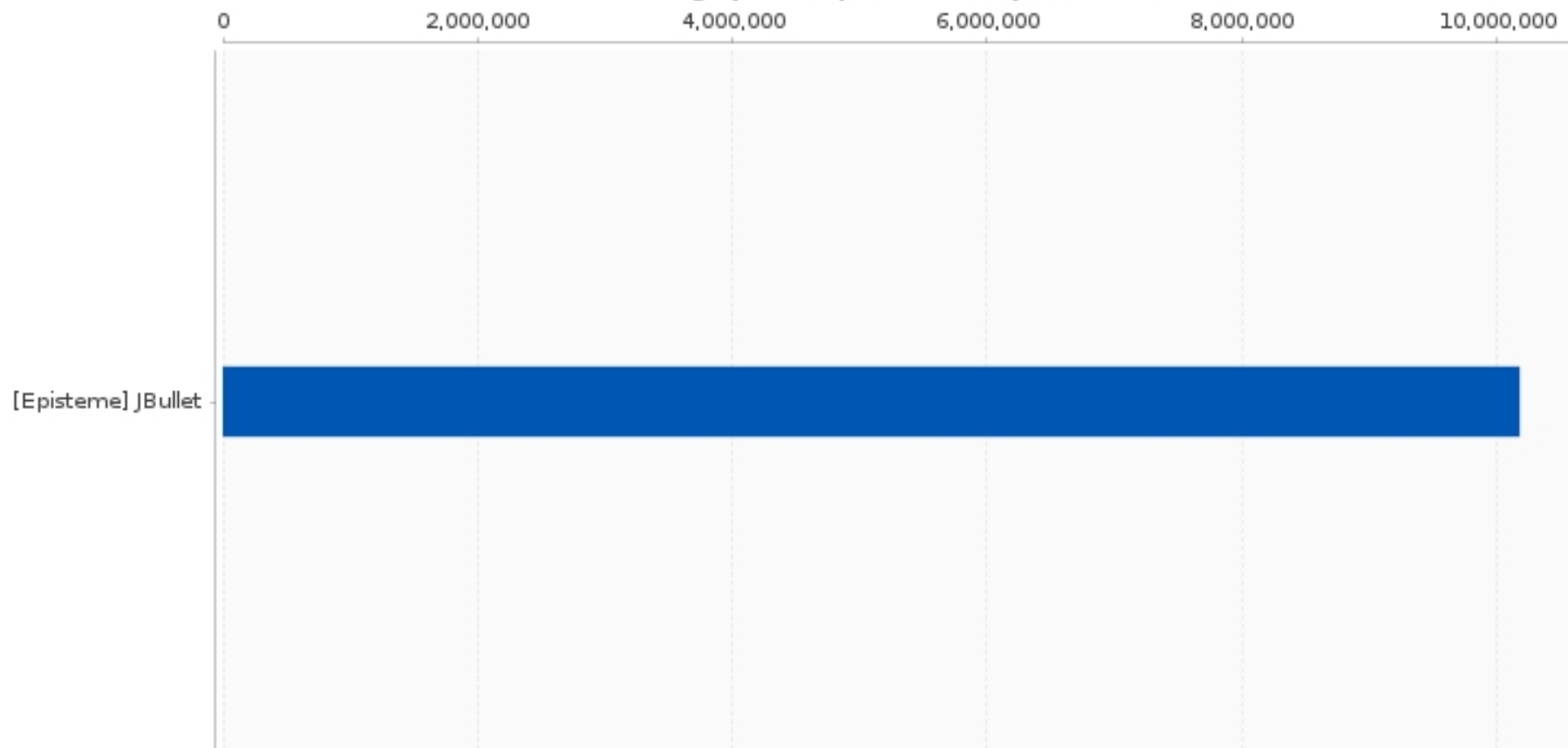
Throughput (Operations per Second)



Quantum Algorithms

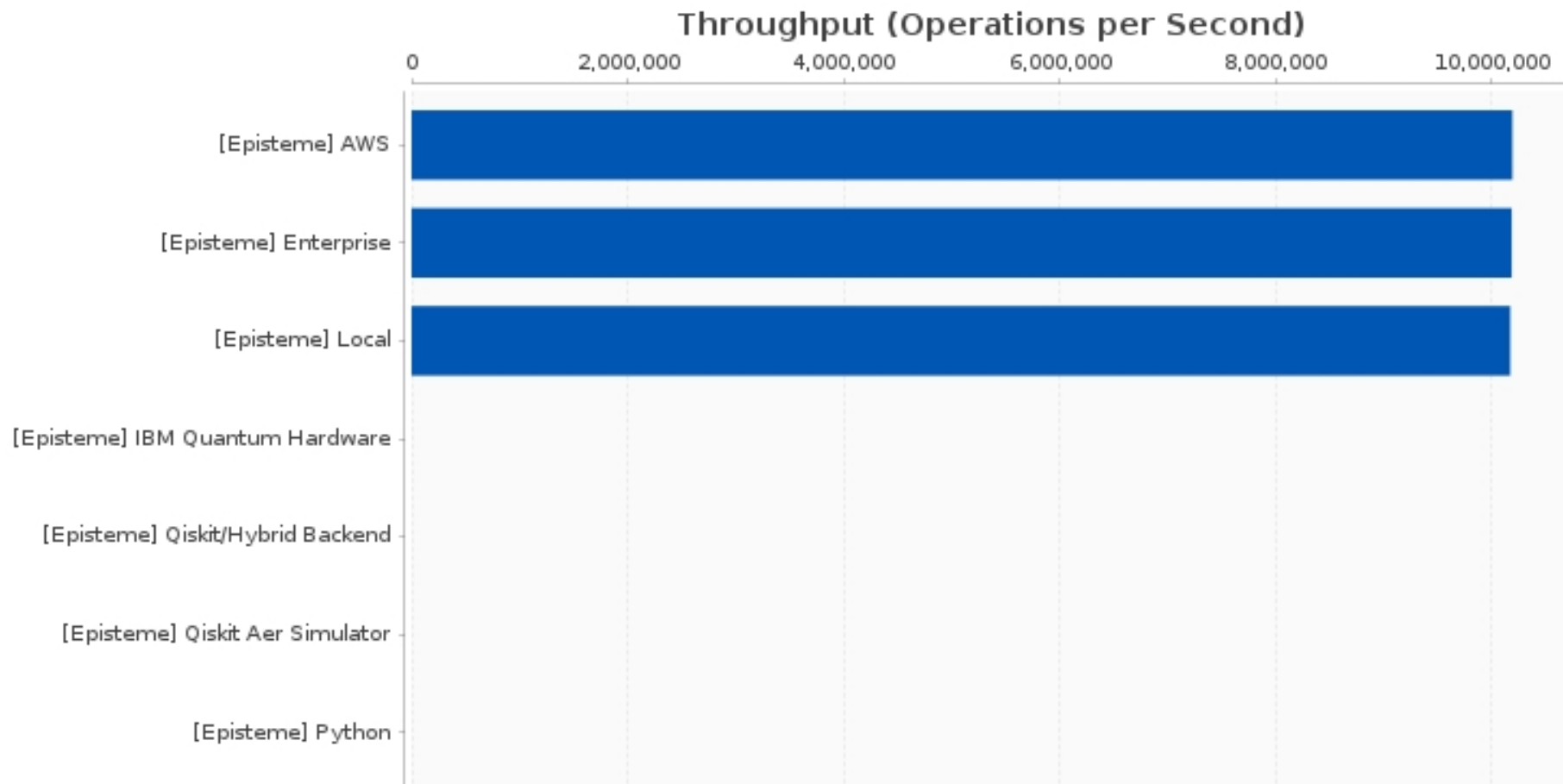
PhysicsEngine | Scaling Performance

Throughput (Operations per Second)



Quantum Simulation (QuEST/Aer)

Quantum Algorithms | Scaling Performance



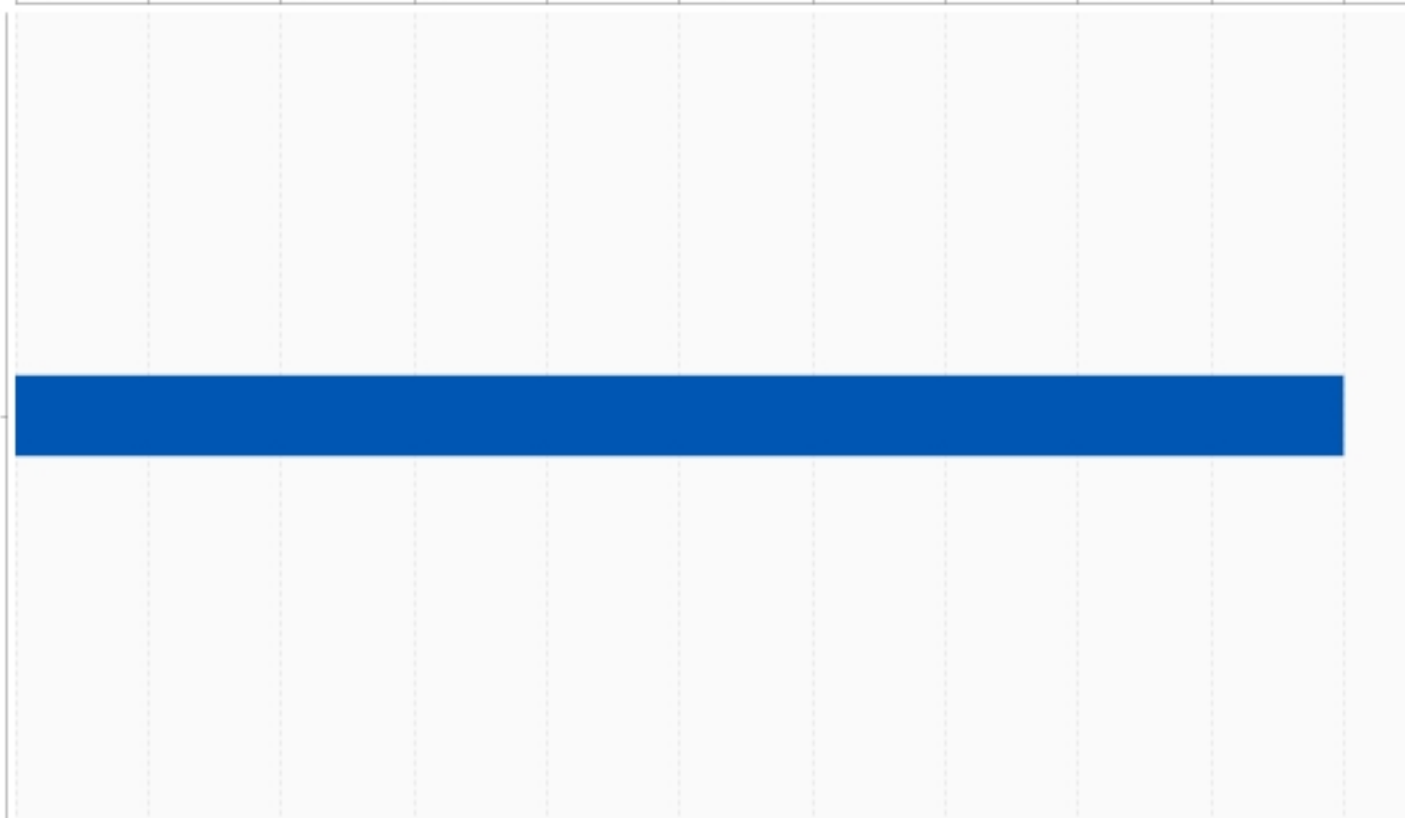
SPH Fluid Dynamics

Quantum Simulation (QuEST/Aer) | Scaling Performance

Throughput (Operations per Second)

0.000 0.001 0.002 0.003 0.004 0.005 0.006 0.007 0.008 0.009 0.010

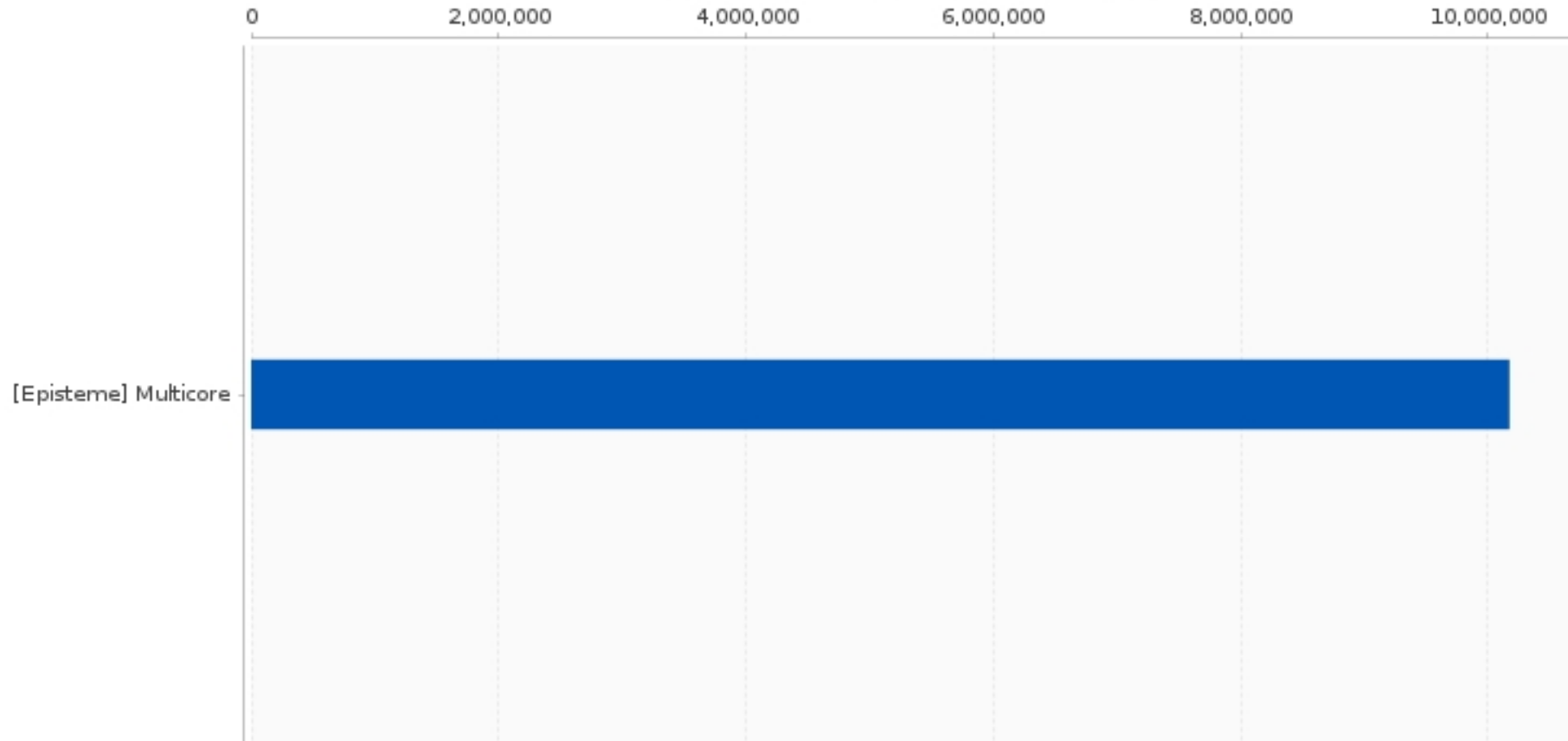
[Episteme Native] QuEST/Qiskit



Self-Consistent Field

SPH Fluid Dynamics | Scaling Performance

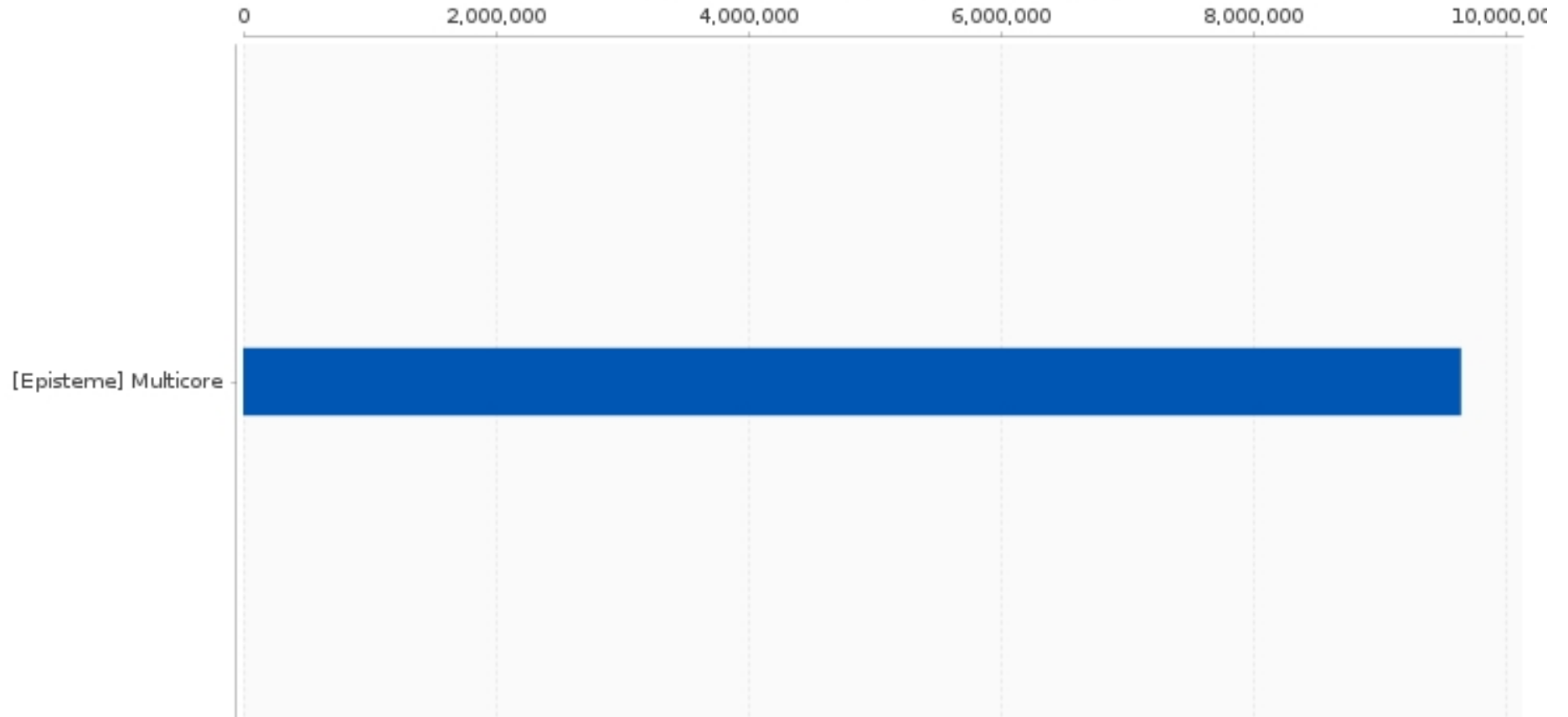
Throughput (Operations per Second)



Signal Processing

Self-Consistent Field | Scaling Performance

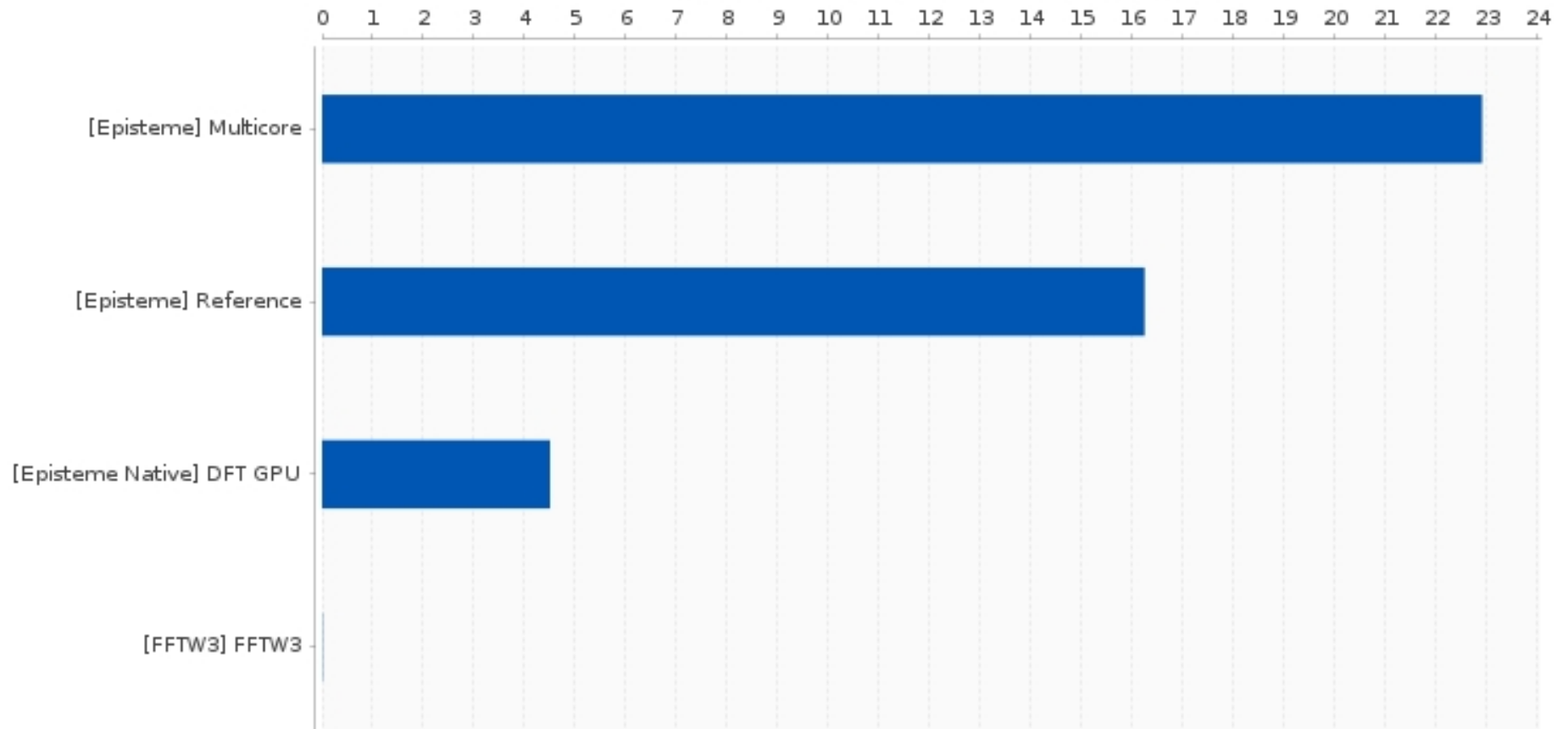
Throughput (Operations per Second)



Social / Geography

Signal Processing | Scaling Performance

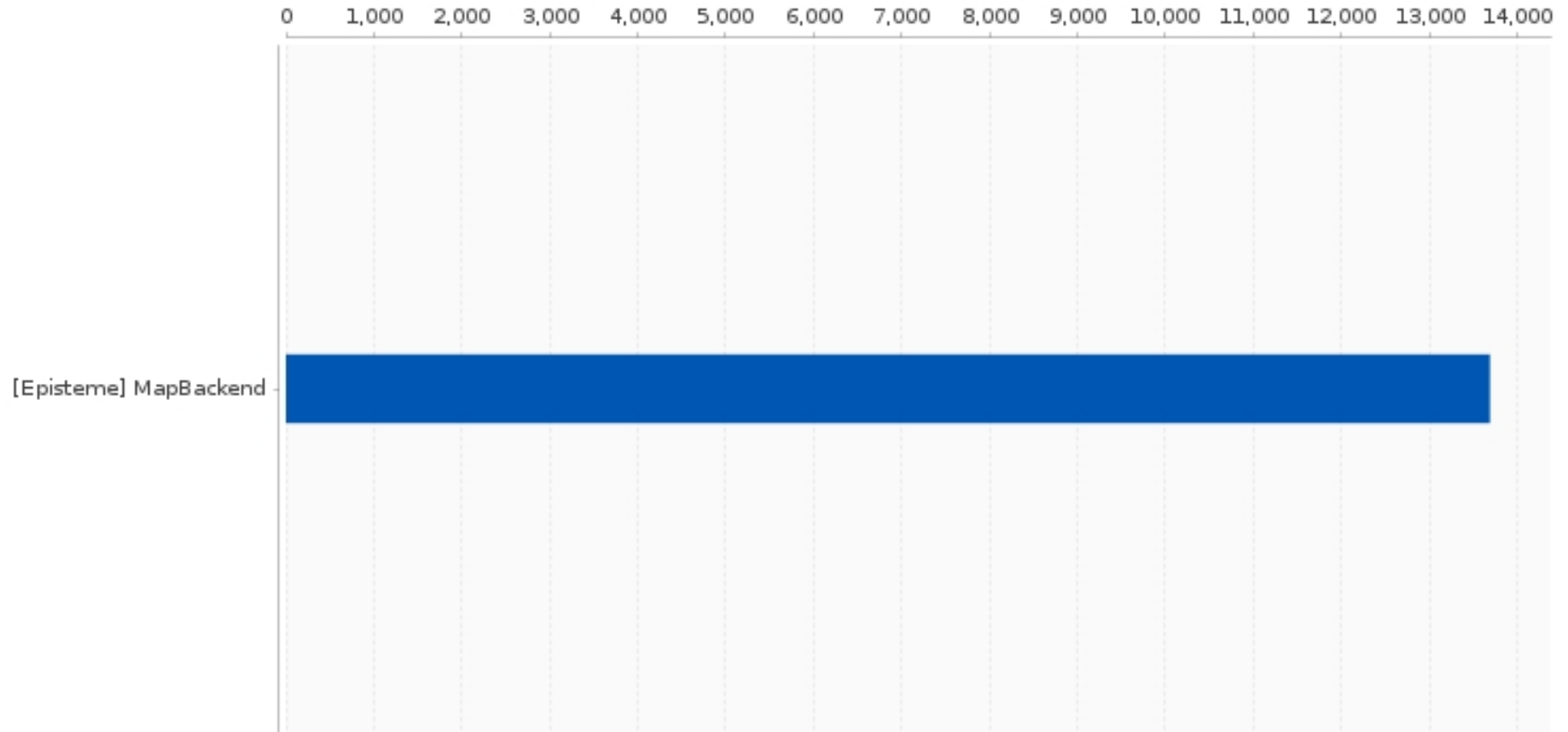
Throughput (Operations per Second)



Tensor Operations

Social / Geography | Scaling Performance

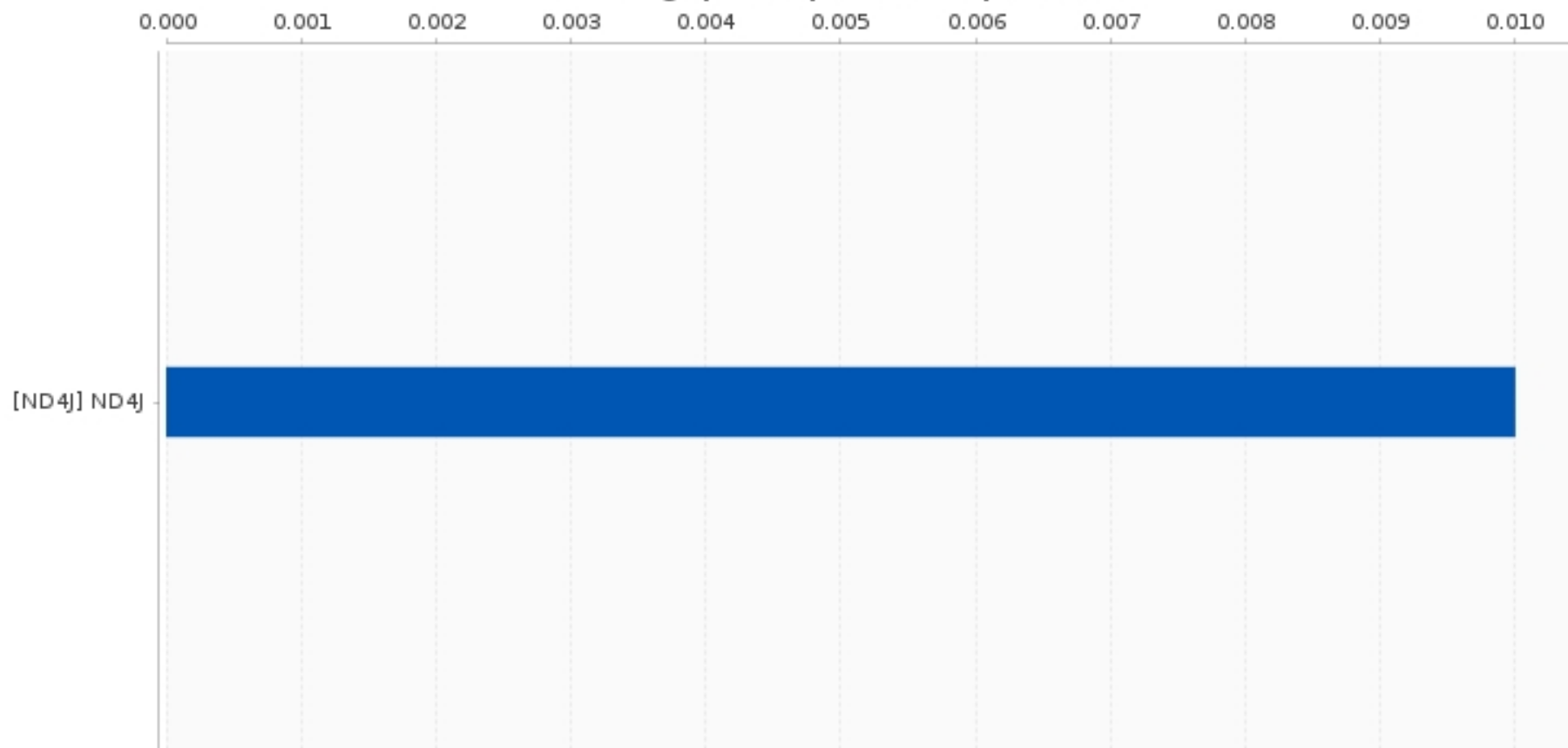
Throughput (Operations per Second)



Visualization

Tensor Operations | Scaling Performance

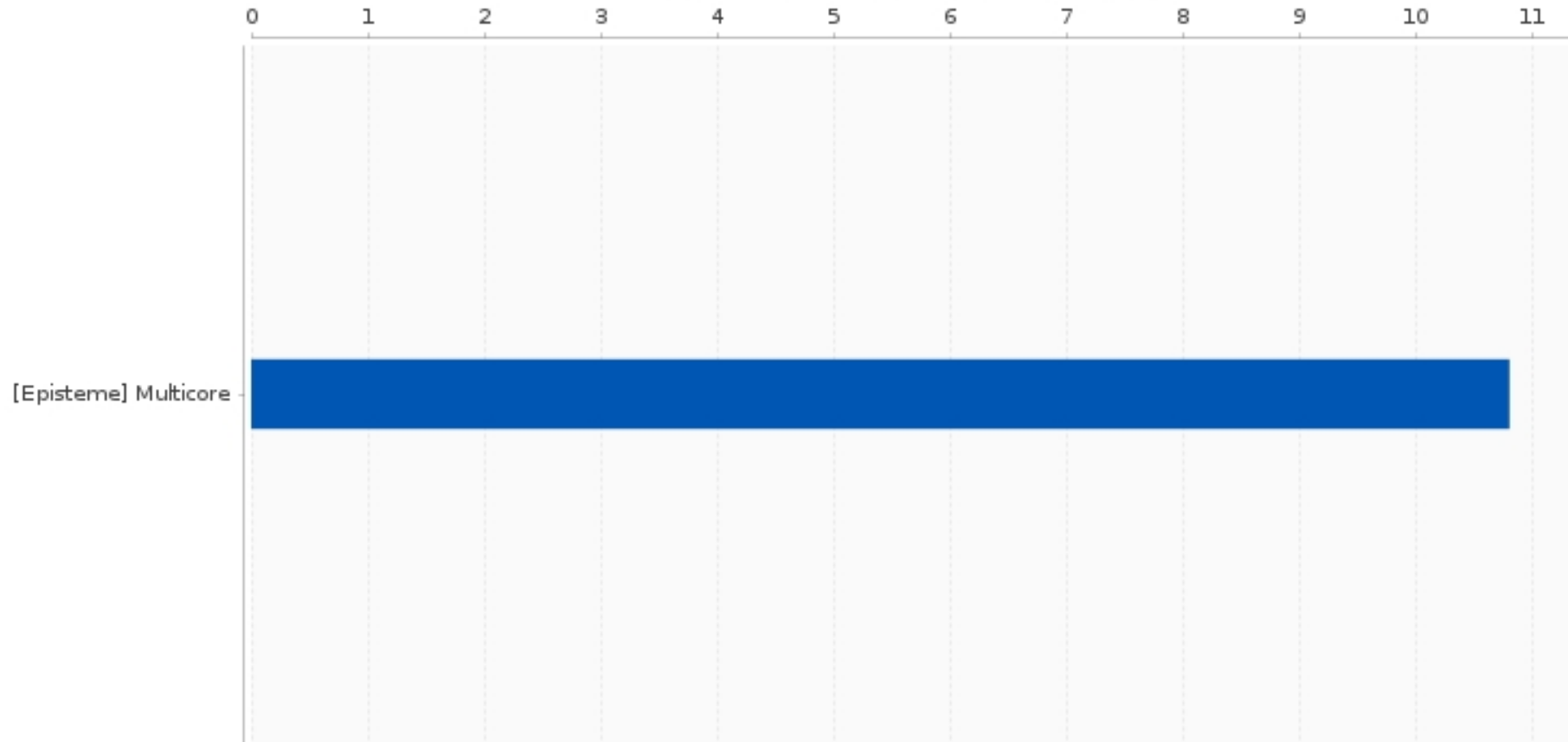
Throughput (Operations per Second)



Wave Equation

Visualization | Scaling Performance

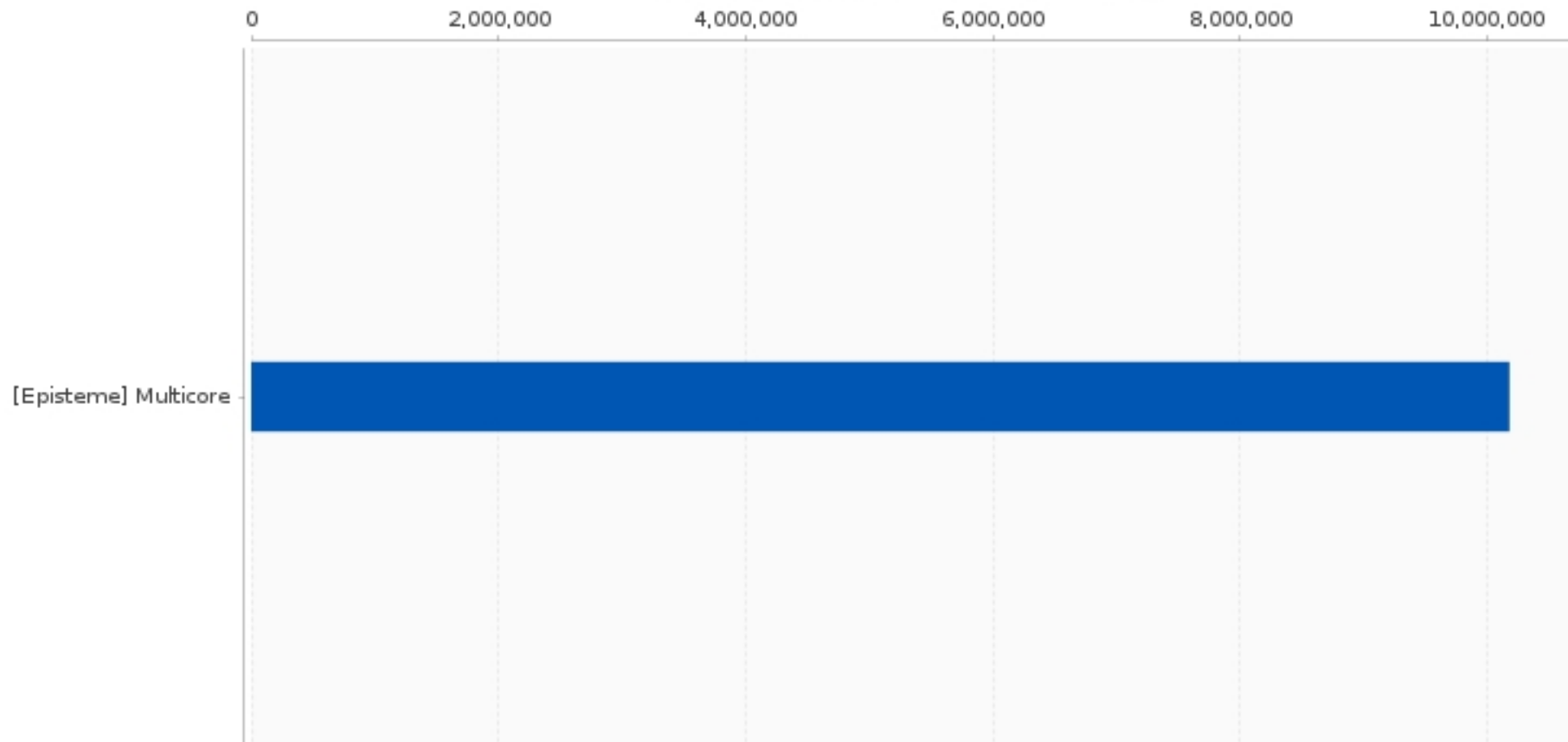
Throughput (Operations per Second)



Zero-Copy Data (Arrow)

Wave Equation | Scaling Performance

Throughput (Operations per Second)



Zero-Copy Data (Arrow) | Scaling Performance

